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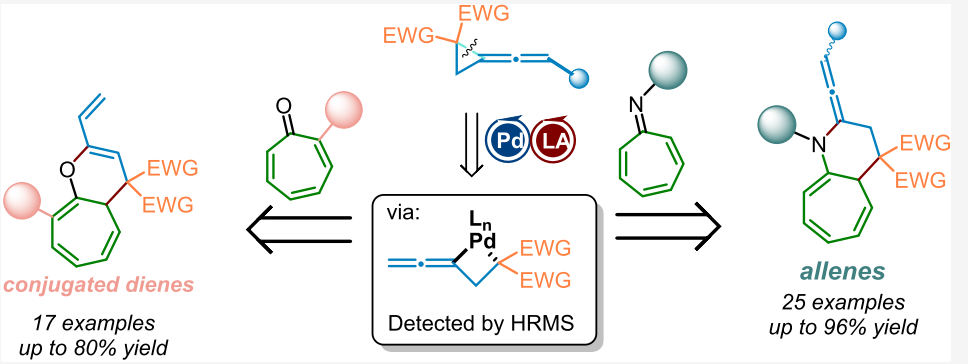
Substrate-Controlled [8 + 3] Cycloaddition of Tropsulfimides and Tropones with Zwitterionic Allenyl Palladium Species Derived from Vinylidenecyclopropane-diesters

Yong-Jie Long, Jia-Hao Shen, Yin Wei,* and Min Shi*

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ABSTRACT: A palladium-catalyzed regioselective [8 + 3] cycloaddition of tropsulfimides and tropones with vinylidenecyclopropane-diesters (VDCP-diesters) has been disclosed in this paper, affording decahydro-1*H*-cyclohepta[*b*]pyridine derivatives bearing an allene moiety or decahydro-1*H*-cyclohepta[*b*]pyran derivatives having a conjugated diene unit in moderate to good yields. The reactions proceed through a zwitterionic allenyl palladium species derived from VDCP-diesters. The substrate scopes have been investigated and the plausible reaction mechanisms have also been proposed according to the previous work, the first captured zwitterionic Pd-allenyl intermediate, and control experiments.

INTRODUCTION

Allenes, recognized as cumulated dienes, and conjugated dienes have been widely used as useful building blocks in natural product synthesis,[1−3] medicinal chemistry,[4] and material science[5−7] as well as fine chemical synthesis.[8−15] Particularly, substances containing allene and conjugated diene subunits have drawn significant attention from organic chemists for the preparation of complex molecular skeletons.[16] Among their synthetic methodologies, the allenyl or propargyl palladium species derived from an oxidative addition of propargyl derivatives bearing a leaving group plays a key role for the rapid preparation of various allene and conjugated diene derivatives (Scheme 1a).[17−23] In general, as for these propargyl derivatives having the large alkyl or aryl substituents located at the leaving group site, the allenic product can be the major one if in the presence of a bidentate phosphine ligand upon nucleophilic addition (Scheme 1a).[24−26] However, when the substituents at the leaving group site are two hydrogen atoms or small alkyl or aryl groups, the conjugated diene can become the major product in the presence of a monodentate phosphine ligand (Scheme 1).[27] The production of an alkynyl compound from nucleophilic substitution of allenyl or propargyl palladium species is relatively rare (Scheme 1). Sometimes, adjusting the nucleophilicity of the employed nucleophile can alter the distribution of allenes and conjugated dienes.[28−31]

On the other hand, the zwitterionic allenyl or propargyl palladium species, as a special type of allenyl or propargyl palladium species, has been seldom seen in organic synthesis although its analogues such as zwitterionic π-allyl palladium species have been widely used in palladium-catalyzed chemical transformations (Scheme 1b).[32−45] In this aspect, zwitterionic allenyl or propargyl copper species generated from acetylenyl ethylene carbonates or carbamates[46−48] have been used in the nucleophilic substitutions for the production of alkynyl or allenyl derivatives (Scheme 1b).[49−51]

In 2021, our group reported the first example on the generation of zwitterionic allenyl or propargyl palladium species from vinylidenecyclopropane-diesters (VDCP-die-

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Scheme 1. (a–c) Previous Work and (d) This Work

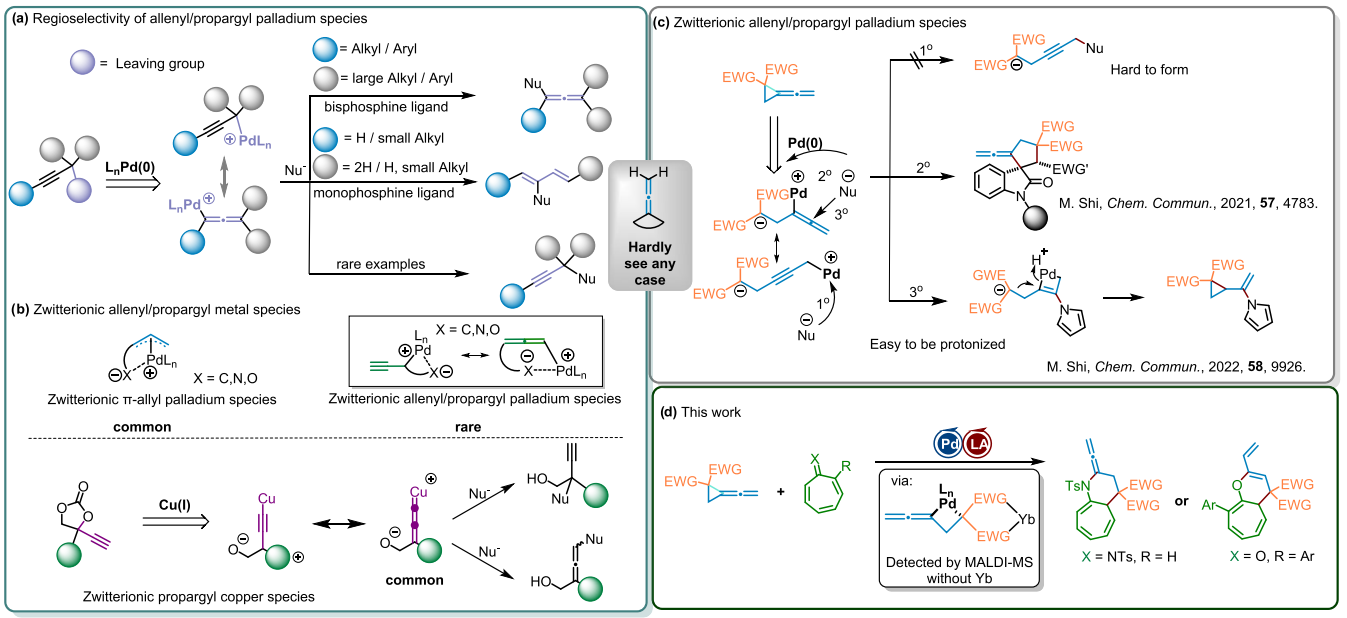
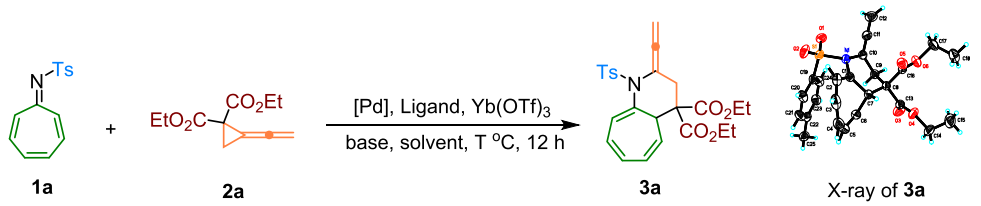


Table 1. Optimization of Reaction Conditions for Tropsulfimide 1a with VDCP 2a



entry[a]	T (°C)	[Pd]	ligand	solvent	base	yield[b] (%)
1	60	Pd_2dba_3	DPEphos	THF	Cs_2CO_3	70
2	60	Pd_2dba_3	BINAP	THF	Cs_2CO_3	32
3	60	Pd_2dba_3	dppf	THF	Cs_2CO_3	57
4	60	Pd_2dba_3	DPEphos	toluene	Cs_2CO_3	56
5	60	Pd_2dba_3	DPEphos	CH_3CN	Cs_2CO_3	27
6	60	Pd_2dba_3	DPEphos	DCE	Cs_2CO_3	trace
7	60	Pd_2dba_3	DPEphos	acetone	Cs_2CO_3	30
8	60	$Pd(PPh_3)_4$	DPEphos	THF	Cs_2CO_3	72
9[c]	60	Pd_2dba_3	DPEphos	THF	Cs_2CO_3	36
10	60	Pd_2dba_3	DPEphos	THF	K_3PO_4	-
11	60	Pd_2dba_3	DPEphos	THF	K_2HPO_4	-
12	60	Pd_2dba_3	DPEphos	THF	KH_2PO_4	-
13	60	Pd_2dba_3	DPEphos	THF	$CsHCO_3$	14
14	60	Pd_2dba_3	DPEphos	THF	DBU	15
15	60	Pd_2dba_3	DPEphos	THF	TEA	43
16[d]	60	Pd_2dba_3	DPEphos	THF	Cs_2CO_3	76
17[e]	**70**	**Pd_2dba_3**	**DPEphos**	**THF**	**Cs_2CO_3**	**92**

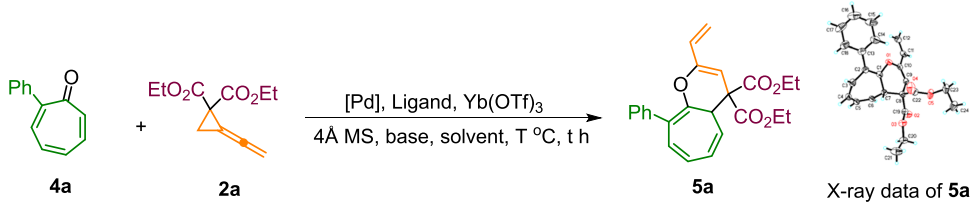
[a]Reaction conditions: a solution of **1a** (0.1 mmol) and **2a** (0.2 mmol), [Pd] (5.0 mol %), ligand (7.5 mol %), $Yb(OTf)_3$ (10.0 mol %), and base (0.1 mmol) in a solvent (1.0 mL) was heated at 60 °C in an oil bath for 12 h. [b]Isolated yield. [c]Without $Yb(OTf)^3$. [d]With 4 Å MS (100 mg). [e]With 4 Å MS (100 mg) and heated to 70 °C in an oil bath.

sters).[52] This newly generated palladium species can undergo a nucleophilic attack or reductive elimination on the Pd-allenyl carbon center or the middle carbon center of allenyl palladium species, affording the corresponding cyclization products or the vinylcyclopropane derivatives (Scheme 1c).[52,53] Therefore, the exploration of new reaction processes and the related regioselectivities is unexploited with regard to this new palladium species.

Tropimide or tropone derivatives are prevalent in natural products and drug-like molecules, and these substances with the skeleton of cyclohepta[*b*]pyridine or cyclohepta[*b*]pyran play a significant role in medicinal chemistry.[54,55] Although there are several reports on the use of tropimides or tropones in methodology,[56−62] the synthetic protocols using them in the production of cyclohepta[*b*]pyridine or cyclohepta[*b*]pyran derivatives are still limited and the exploration of a new

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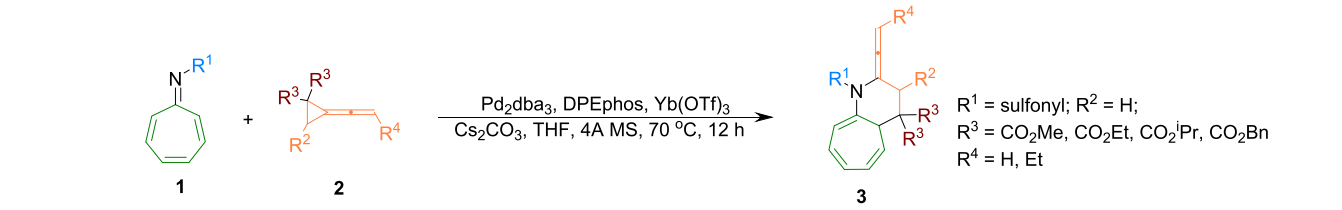
Table 2. Optimization of Reaction Conditions for Tropone 4a with VDCP 2a

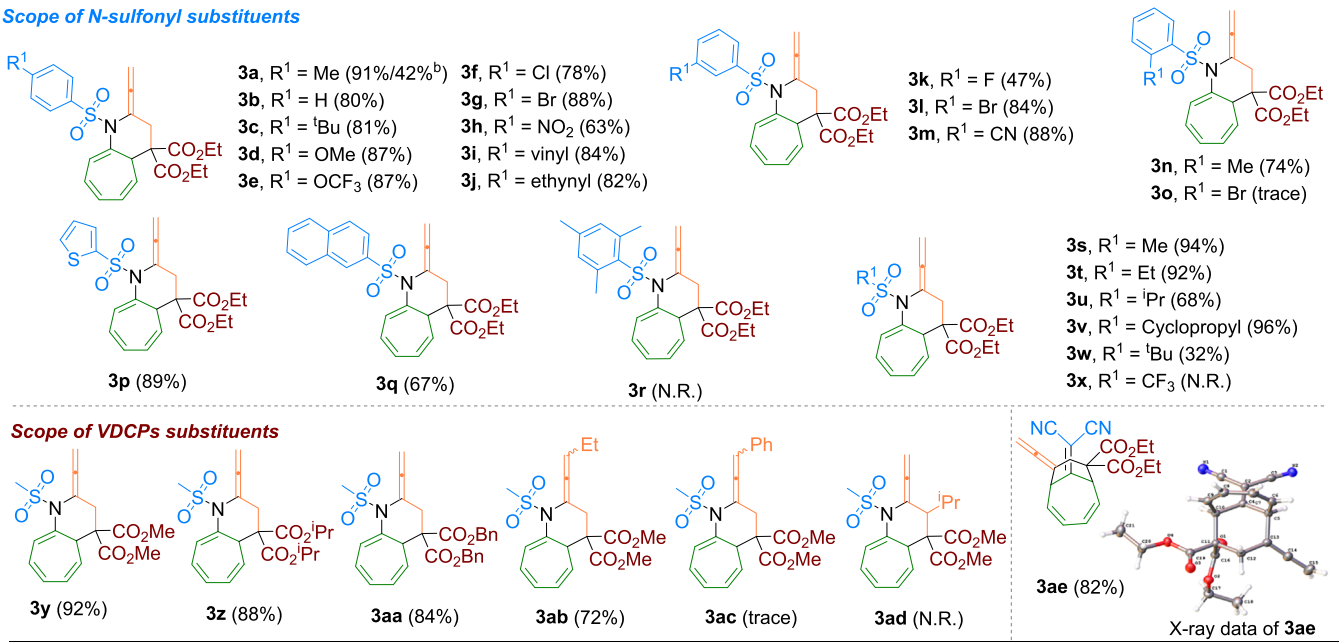


entry[a]	T (°C)	t (h)	ligand	solvent	base	yield[b] (%)
1	45	12	DPEphos	THF	Cs_2CO_3	32
2	45	24	DPEphos	THF	Cs_2CO_3	51
3	50	24	DPEphos	THF	Cs_2CO_3	67
4	50	16	DPEphos	THF	Cs_2CO_3	53
5	50	36	DPEphos	THF	Cs_2CO_3	67
6	55	24	DPEphos	THF	Cs_2CO_3	66
7	50	24	DPEphos	toluene	Cs_2CO_3	16
8	50	24	DPEphos	1,4-dioxane	Cs_2CO_3	52
9	50	24	DPEphos	DMF	Cs_2CO_3	48
10	50	24	BINAP	THF	Cs_2CO_3	trace
11	50	24	dtbpf	THF	Cs_2CO_3	39
12	50	24	DPEphos	THF	TEA	-
13	50	24	DPEphos	THF	DBU	-
14	50	24	DPEphos	THF	K_3PO_4	-

[a]Reaction conditions: a solution of **4a** (0.1 mmol) and **2a** (0.2 mmol), [Pd] (5.0 mol %), ligand (7.5 mol %), $Yb(OTf)_3$ (10.0 mol %), 4 Å MS (100 mg), and base (0.1 mmol) in a solvent (1.0 mL) in an oil bath. [b]Isolated yield.

Scheme 2. Substrate Scope of [8 + 3] Cycloaddition between Tropsulfimides 1 and VDCPs 2[a]

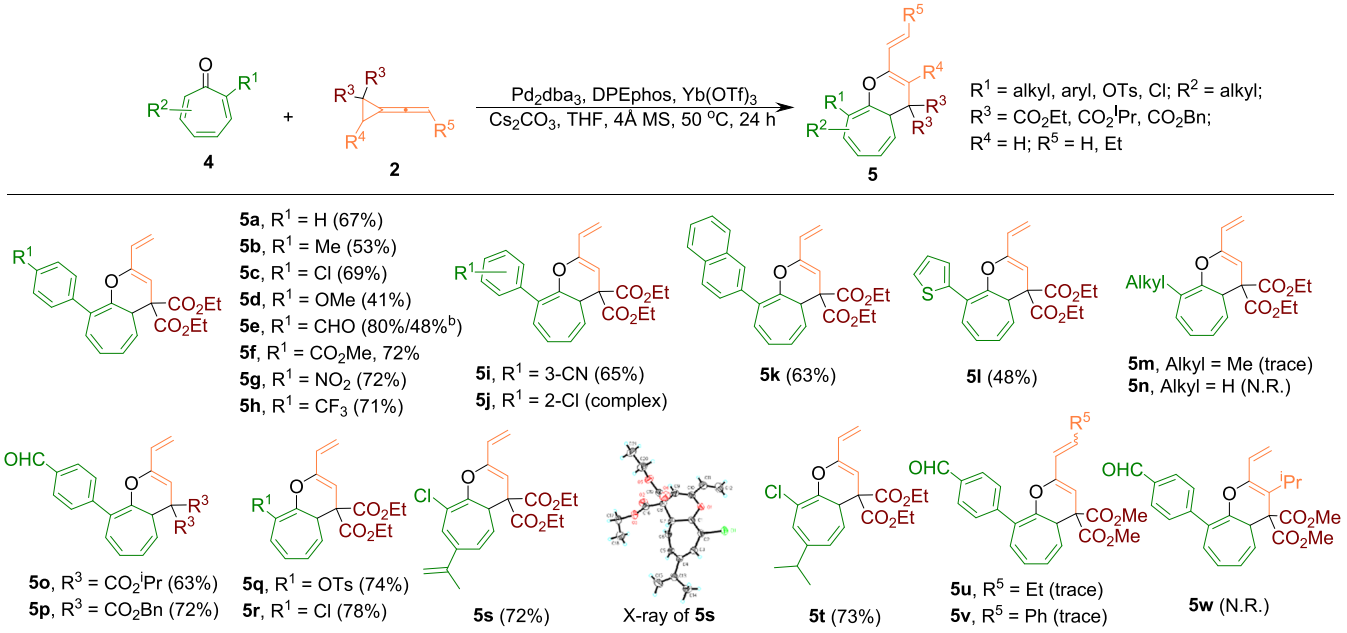




[b]Scale-up reaction: **1a** (259 mg, 1.0 mmol), **2a** (0.35 mL, 2.0 mmol), Pd_2dba_3 (23 mg, 2.5 mol %), DPEphos (40 mg, 7.5 mol %), $Yb(OTf)_3$ (62 mg, 10.0 mol %), Cs_2CO_3 (325 mg, 1.0 mmol), 4 Å MS (1.0 g), and THF (10.0 mL) heated at 70 °C in an oil bath for 12 h. [a]Reaction conditions: a solution of **1** (0.1 mmol) and **2** (0.2 mmol), Pd_2dba_3 (2.5 mol %), DPEphos (7.5 mol %), $Yb(OTf)_3$ (10.0 mol %), 4 Å MS (100 mg), and Cs_2CO_3 (0.1 mmol) in THF (1.0 mL) was heated at 70 °C in an oil bath for 12 h, isolated yield.

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Scheme 3. Substrate Scope of [8 + 3] Cycloaddition between Tropones 4 and VDCPs 2[a]



[b]Scale-up reaction: **4e** (210 mg, 1.0 mmol), **2a** (0.35 mL, 2.0 mmol), Pd_2dba_3 (23 mg, 2.5 mol %), DPEphos (40 mg, 7.5 mol %), $Yb(OTf)_3$ (62 mg, 10.0 mol %), Cs_2CO_3 (325 mg, 1.0 mmol), 4 Å MS (1.0 g), and THF (10.0 mL) heated at 50 °C in an oil bath for 24 h. [a]Reaction conditions: a solution of **4** (0.1 mmol) and **2** (0.2 mmol), Pd_2dba_3 (2.5 mol %), DPEphos (7.5 mol %), $Yb(OTf)_3$ (10.0 mol %), 4 Å MS (100 mg), and Cs_2CO_3 (0.1 mmol) in THF (1.0 mL) was heated at 50 °C in an oil bath for 24 h, isolated yield.

method to generate such skeletons is highly desirable. In view of the aforementioned research circumstances and our previous work on zwitterionic allenyl or propargyl palladium species derived from vinylidenecyclopropane-diesters,[52,53] we attempted to investigate the cycloaddition reaction between this zwitterionic palladium species and tropsulfimides or tropones to clarify the structures of the obtained cycloadducts. In this context, we wish to report our findings (Scheme 1d).

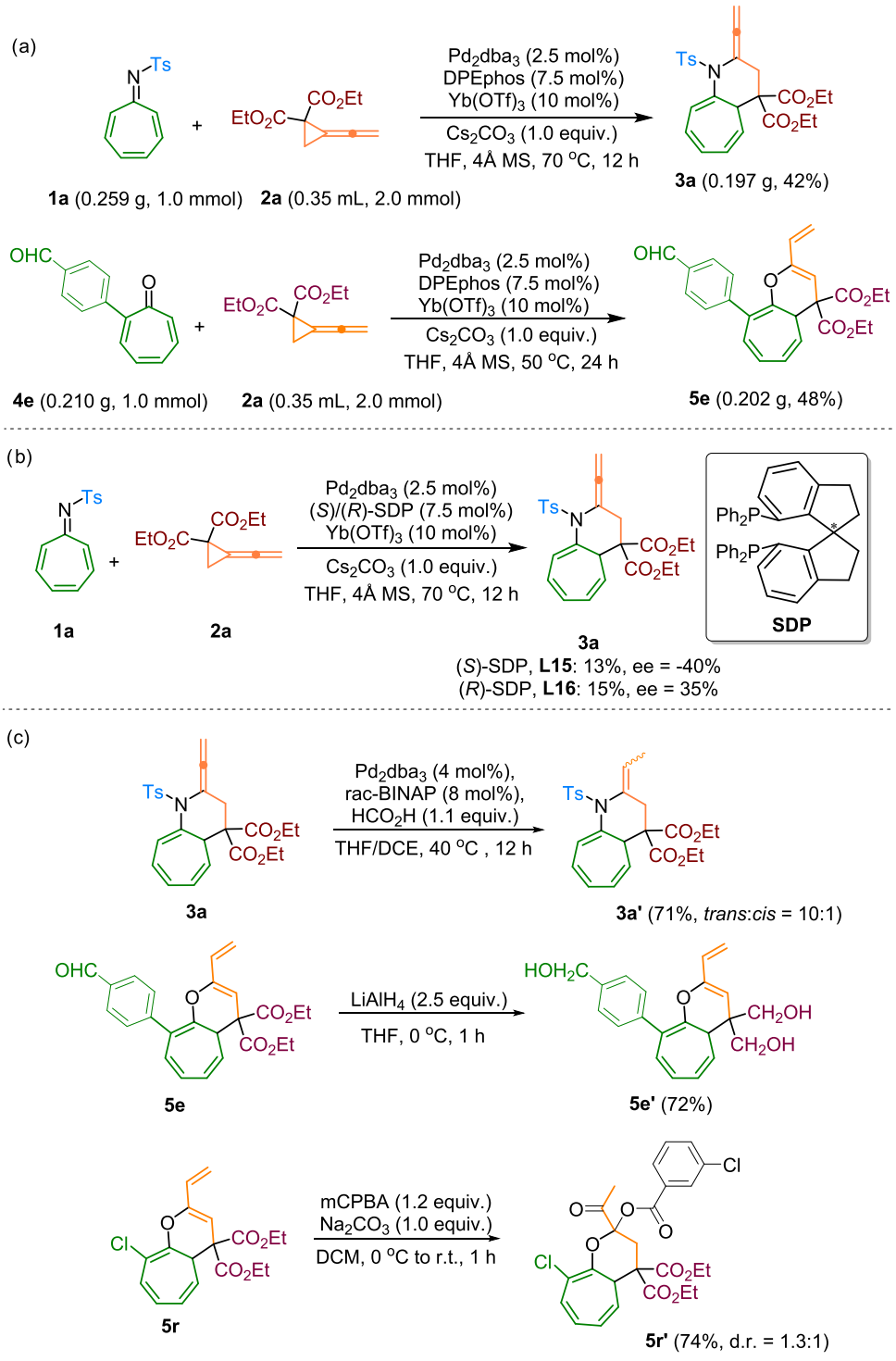
RESULTS AND DISCUSSION

In our initial experimental examination, tropsulfimide **1a** and VDCP-diester **2a** were used as model substrates to explore the reaction outcomes and subsequently optimize the reaction conditions using Pd_2dba_3 as a palladium catalyst, DPEphos as a ligand, $Yb(OTf)_3$ as a Lewis acid additive, and Cs_2CO_3 as an inorganic base additive in tetrahydrofuran (THF) at 60 °C for 12 h. Exhilaratingly, the corresponding [8 + 3] cycloaddition product **3a** bearing an allenyl moiety was obtained as a single product in the yield of 70% (Table 1, entry 1). Its structure has been unequivocally determined by X-ray diffraction. The ORTEP drawing is shown in Table 1, and the CIF data are summarized in the Supporting Information. The use of BINAP and dppf as ligands gave **3a** in lower yields (32% yield with BINAP and 57% yield with dppf) than that of using DPEphos (Table 1, entries 2 and 3). The examination of solvent effects such as toluene, CH_3CN, 1,2-dichloroethane (DCE), and acetone revealed that THF is the solvent of choice, giving **3a** in higher yields (Table 1, entries 4−7). Changing the palladium source from Pd_2dba_3 to $Pd(PPh_3)_4$ did not affect the result significantly, giving **3a** in 72% yield (Table 1, entry 8). Removing $Yb(OTf)_3$ from the catalytic system obviously lowered the yield of **3a** to 36% (Table 1, entry 9). In consideration of the stability on the anionic site of zwitterionic palladium species, which was brought about from the base,[63] we examined the base effect using the other inorganic bases such as K_3PO_4, K_2HPO_4, KH_2PO_4, and $CsHCO_3$ or organic bases such as DBU and triethylamine (TEA); Cs_2CO_3 still gave the best result (Table 1, entries 10−15). To get rid of the ambient moisture, 4 Å MS (100 mg) was added into the reaction system and we identified that the reaction produced **3a** in 76% yield under otherwise identical conditions (Table 1, entry 16). Moreover, raising the reaction temperature to 70 °C afforded **3a** in 92% yield, which served as the optimal reaction conditions for this palladium-catalyzed transformation (Table 1, entry 17).

Next, instead of tropsulfimide **1a**, we also examined the reaction of tropone **4a** with **2a** under similar conditions aforementioned and found that carrying out the reaction at 45 °C provided another [8 + 3] cycloaddition product **5a** having a vinyl moiety in 32% yield within 12 h (Table 2, entry 1). Its structure has been also unambiguously identified by X-ray diffraction, and the ORTEP drawing is shown in Table 2. Extending the reaction time to 24 h improved the yield of **5a** to 51% (Table 2, entry 2), and elevating the reaction temperature to 50 °C gave **5a** in 67% yield within 24 h and 53% yield within 16 h, which was similar to that of **3a** (Table 2, entries 3 and 4). However, prolonging the reaction time or raising the temperature further had a slight effect on the reaction outcome (Table 2, entries 5 and 6). The examination of solvent effects disclosed that using toluene, 1,4-dioxane, or *N*,*N*-dimethylformamide (DMF) as the solvent gave no improvement of the reaction outcome (Table 2, entries 7−9) and the use of BINAP or dtbpf as a ligand did not afford **5a** in a better result (Table 2, entries 10 and 11). None of the desired product **5a** was formed upon using TEA, DBU, or K_3PO_4 as a base under the standard conditions (Table 2, entries 12−14). Overall, the optimal conditions for the production of **5a** include carrying

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Scheme 4. (a) Scale-up Reaction, (b) Asymmetric Version of 1a with 2a, and (c) Synthetic Transformation

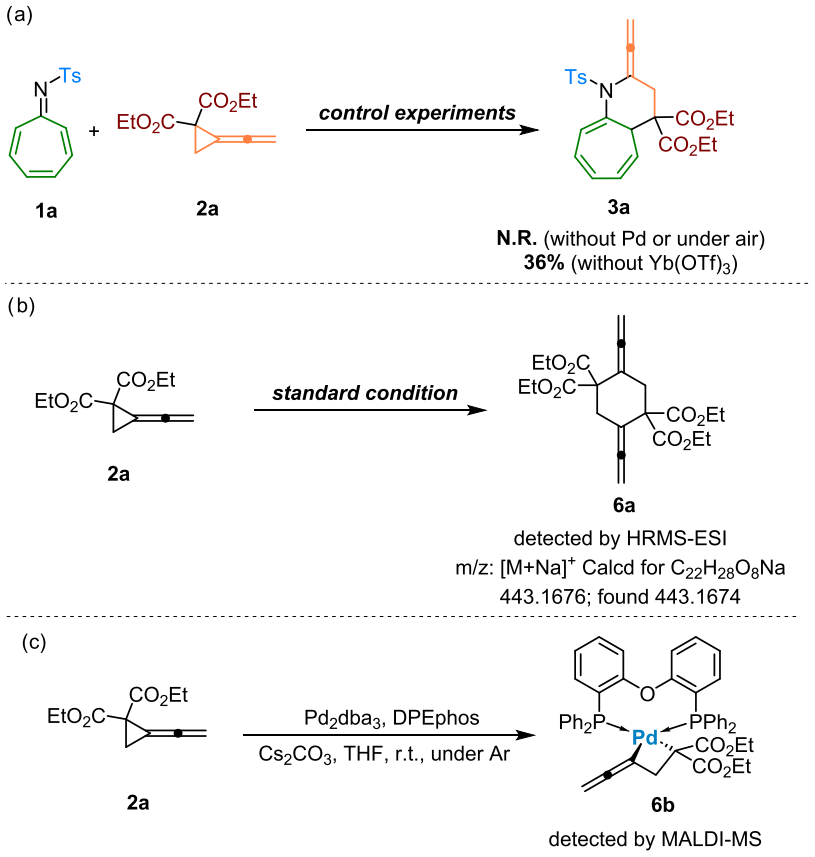


out the reaction in THF at 50 °C for 24 h in the presence of Cs_2CO_3.

First, the substrate scope for the production of **3** was examined and the results are shown in Scheme 2. A wide range of tropsulfimides **1** reacted with the zwitterionic allenyl palladium species generated from VDCP-diesters **2** smoothly, affording the desired cyclized products **3** in moderate to good yields. Regardless of whether an electron-withdrawing or -donating substituent was introduced at the *para*-position of the benzene ring linked to sulfimides, the target products **3a**–**3j** were obtained in 63 to 91% yields, suggesting that the electronic property of these substituents did not have significant impact on the reaction outcome. The substituent at the *meta*-position of the benzene ring also gave the corresponding cycloadducts **3k**–**3m** in 47 to 88% yields. It should be noted that introducing a methyl group at the *ortho*-position of the benzene ring delivered **3n** in 74% yield; however, introducing a bromine atom at the *ortho*-position did not afford **3o**, presumably due to the steric effect.

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Scheme 5. (a–c) Control Experiments and Investigation on the Reactive Intermediates



Tropsulfimides **1p** and **1q** having a heterocycle or a fused aromatic ring were also tolerated, giving the desired products **3p** and **3q** in 89 and 67% yields, respectively. A sterically bulky aromatic ring such as a mesitylene group probably impaired the cycloaddition reaction and did not give the corresponding product **3r** under identical conditions. Meanwhile, introducing a variety of alkyl groups to sulfimides **1** afforded the corresponding products **3s–3v** in moderate to good yields ranging from 68–96%. In the case of sulfimide **1w**, the desired cycloadduct **3w** was obtained in 32% yield probably due to the steric effect as well. However, using sulfimide **1x** as a substrate, in which a strongly electron-withdrawing CF_3 group was introduced, did not furnish the cyclized product **3x** perhaps due to the electronic effect leading to the weak *N*-nucleophilicity of sulfimide **1x**.

Furthermore, a variety of VDCP-diesters were compatible to this reaction, affording the desired [8 + 3] cycloaddition products **3y–3ab** in good to excellent yields ranging from 72–92%. However, the reactions did not proceed smoothly when the terminal position of the allene moiety in **2** was changed from the ethyl to a phenyl group or an isopropyl group was added on the cyclopropane ring. The corresponding products **3ac** and **3ad** were not formed under the standard conditions. Remarkably, substrate **1ae** containing a malononitrile moiety instead of a sulfimide unit, which was prepared upon condensation between tropone and malononitrile, underwent a [6 + 3] cycloaddition reaction pathway rather than a [8 + 3] cycloaddition process probably due to the steric effect, furnishing the corresponding product **3ae** in 82% yield. Its structure has been clarified by X-ray diffraction, and the related CIF data are depicted in the Supporting Information.

Next, the substrate scope for the production of **5** using tropones as substrates was subsequently explored and the results are exhibited in Scheme 3. A variety of tropones **4** were employed to react with VDCP-diesters **2**, providing the products **5** in moderate to good yields. Introducing various substituents at the *para*-position of the benzene ring located at the 2-position of the cycloheptatriene ring furnished the corresponding products **5a–5h** in the yields of 41–80%, also rendering that the electronic property of these substituents did not have significant impact on the reaction outcome. In addition, tropone **4i** bearing a cyano group at the *meta*-position of the benzene ring gave the desired product **5i** in 48% yield. Introducing a chlorine atom at the *ortho*-position of the benzene ring such as tropone **4j** did not give the desired product **5j** perhaps due to the steric effect. Moreover, tropones **4k** and **4l** containing a fused aromatic ring and a heterocycle could also be tolerated, furnishing the corresponding products **5k** and **5l** in 63 and 48% yields, respectively. Nevertheless, tropone **4m** only with a methyl group and tropone **4n** without any substituent could not react with **2a** to give the corresponding cycloadducts, presumably due to their low reactivity with the zwitterionic allenyl palladium species generated from VDCP-diesters. Meanwhile, changing the ester group in VDCP-diester **2** from the ethyl group to an isopropyl group or benzyl group smoothly delivered the corresponding cycloadducts **5o** and **5p** in 63 and 72% yields, respectively. The other electron-withdrawing groups such as OTs or Cl located at the 2-position of the cycloheptatriene ring were also tolerated, giving the desired products **5q** and **5r** in the yields of 74 and 78%, respectively. Tropones **4s** and **4t**, derived from hinokitiol, a natural product, also had excellent

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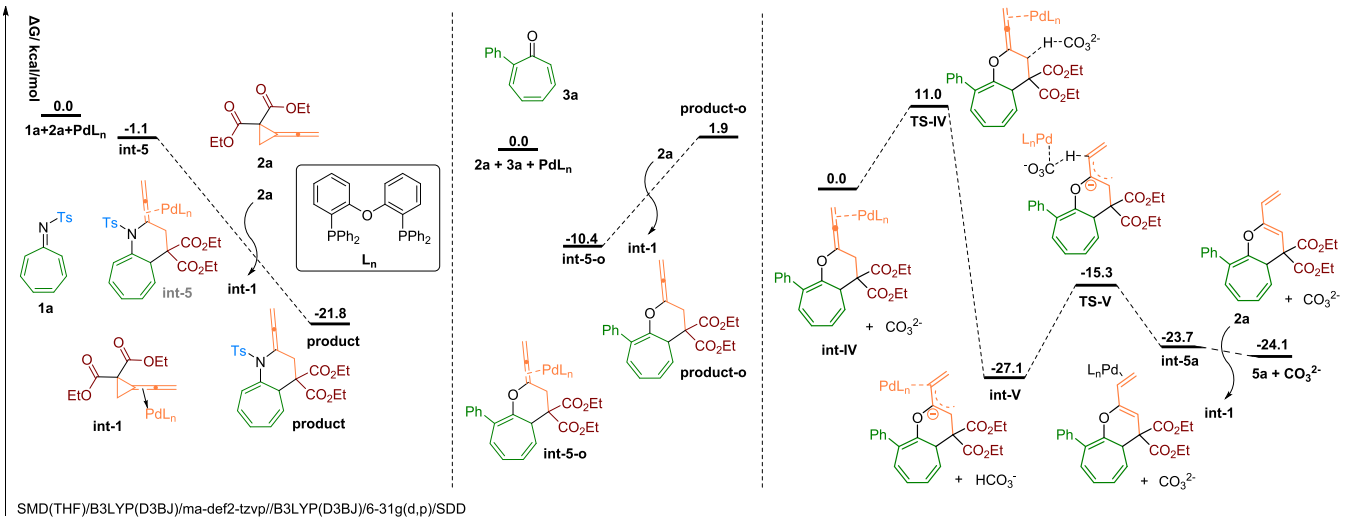


Figure 1. DFT calculation on the rearrangement reaction.

reactivities with the zwitterionic allenyl palladium species to afford the corresponding products **5s** and **5t** in the yields of 72 and 73%, respectively. The structure of **5s** has been also clarified by X-ray diffraction and its ORTEP drawing is shown in Scheme 3. Nevertheless, further examination revealed that introducing an ethyl group or a phenyl group on the terminal side of the allenyl moiety and adding an isopropyl group on the cyclopropane ring did not afford the corresponding products **5u**−**5w** perhaps due to the steric effect.

To illustrate the practicability of these reactions in organic synthesis, a large-scale synthesis of **3a** or **5e** was performed by using **1a** (259 mg, 1.0 mmol) or **4e** (0.210 g, 1.0 mmol) to react with **2a** (0.35 mL, 2.0 mmol) under the standard conditions, affording **3a** or **5e** in 42 or 48% yield, respectively (Scheme 4). The discrepancy between a template reaction and a large-scale synthesis is presumably due to the different thermal, anaerobic and heterogeneous experimental conditions. Likewise, the catalytic asymmetric version of this cycloaddition was also investigated (see Scheme S4 in the Supporting Information for more details), and we found that the use of chiral phosphine ligands **L15** and **L16** delivered **3a** in 13 and 15% yields, respectively, along with −40 and 35% ee values (Scheme 4b). The slight discrepancy is probably due to the experimental error or the partial decomposition of **3a** during the reaction. On the other hand, we have also tried the synergistic catalysis by adding a chiral ammonium salt as reported previously;[64] however, no better result could be obtained (see Scheme S4 in the Supporting Information).

Furthermore, the transformations of the obtained cycloadducts have been performed. The palladium-catalyzed reduction of **3a** in formic acid afforded **3a′** in 71% yield as a *trans* and *cis*-isomeric mixture and the transformation of **5e** was achieved through the reduction with $LiAlH_4$ in THF, affording triol product **5e′** in 72% yield (Scheme 4c). In addition, the oxidation of **5r** was realized under the conditions of *m*-CPBA and Na_2CO_3 in dichloromethane (DCM) at room temperature, giving the corresponding product **5r′** in 74% yield along with 1.3:1 d.r. value (for the plausible reaction mechanism, see Scheme S3 in the Supporting Information) (Scheme 4c).

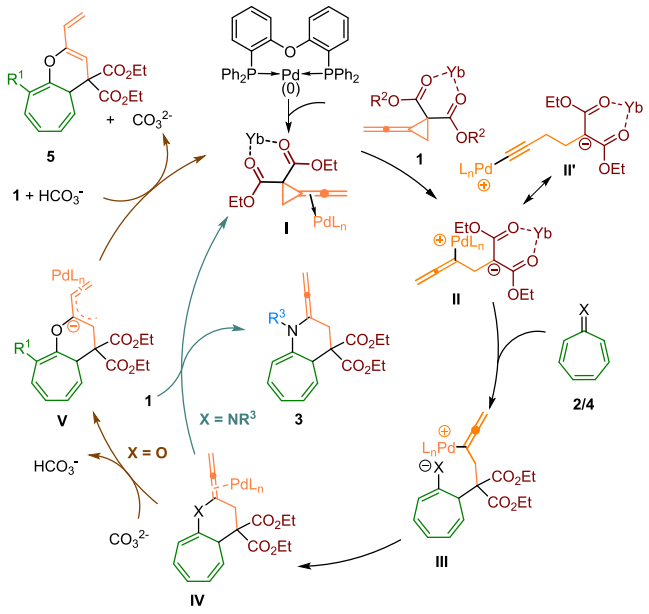
To investigate the reaction mechanism of these cycloaddition reactions, several control experiments were carried out using **1a** to react with **2a** under the standard conditions. As shown in Scheme 5, a, no reaction occurred in the absence of the palladium catalyst or under an ambient atmosphere, while Lewis acid such as $Yb(OTf)_3$ was not an prerequisite catalyst, rendering that these reactions were Pd-catalyzed chemical transformations. On the other hand, VDCP dimer **6a** was detected on the basis of HRMS under the standard conditions upon only using **2a** as the reactant (Scheme 5b). Moreover, allenyl palladium complex **6b**, prepared from Pd_2dba_3, DPEphos, and VDCP-diester **2a**, was observed by matrix-assisted laser desorption ionization mass spectrometry (MALDI-MS), providing the first experimental evidence on the formation of a zwitterionic allenyl palladium species and suggesting that the zwitterionic palladium species might be derived from the oxidative addition of the palladium catalyst with the cyclopropane ring in VDCP-diesters (Scheme 5c).

To further understand the substrate-induced selectivity, density functional theory (DFT) calculations of the rearrangement part were performed (Figure 1). All calculations have been performed at the SMD(THF)/B3LYP-D3BJ/ma-def2TZVP/SDD//B3LYP-D3BJ/6-31G(d,p)/SDD level with the Gaussian 16 program. An evident difference was shown between the dissociation of tropsulfimide and tropone that the former was an exothermic process to form the **product** and regenerate the key intermediate **int-1**, while the latter was an endothermic process to form **product-o**. Moreover, the rearrangement of the allenyl product as **int-V** to the conjugated diene product as **2a** was exothermic, relatively elucidating the substrate-induced selectivity of the reaction.

On the basis of the above experimental results and the previously reported processes, plausible mechanisms for the [8 + 3] cycloaddition of tropsulfimides and tropone derivatives with zwitterionic allenyl palladium species, which can also isomerize to a zwitterionic propargyl palladium species, are proposed in Scheme 6. A coordination between the palladium(0) complex and VDCP-diester **2a** formed complex **I** and induced an oxidative addition to take place at the electron-deficient C−C site of the cyclopropyl unit to form intermediate **II** that can isomerize to zwitterionic propargyl palladium species **II′**, which underwent the cycloaddition reaction with tropsulfimide **1** or tropone derivative **4** to afford intermediate **III**. Since then, intermediate **IV** was generated via a chair-form X^--leading nucleophilic attack. When X = NSO_2R, a ligand exchange caused by **1** afforded the desired product **3** and regenerated the intermediate **I**. When X = O, a proton

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Scheme 6. Plausible Reaction Mechanism



transfer driven by the carbonate anion would occur rather than a ligand exchange, producing the allyl anionic intermediate **V**. Through a proton-transfer shuttle, the protonation of **V** on another side of the allyl anion with the bicarbonate anion could afford product **5** and the ligand exchange with **1** would regenerate the Pd complex **I** for the next catalytic cycle.

CONCLUSIONS

In summary, a novel and regioselective [8 + 3] cycloaddition of tropsulfimides and tropone derivatives with zwitterionic palladium intermediates generated from VDCP-diesters has been realized, providing decahydro-1*H*-cyclohepta[*b*]pyridine derivatives bearing an allene unit or decahydro-1*H*-cyclohepta-[*b*]pyran derivatives containing a conjugated diene moiety in moderate to good yields depending on the C = X (X = N or O) bond in tropone derivatives. Their substrate scopes have been investigated, and the key zwitterionic palladium intermediate was verified through MALDI mass spectroscopy for the first time. The plausible reaction mechanisms have been proposed. Further investigations on the synthesis of biologically active cycloheptatriene derivatives are undergoing in our laboratory.

EXPERIMENTAL SECTION

General Information. Melting points were determined on a digital melting point apparatus and temperatures were uncorrected. Proton nuclear magnetic resonance (^{1}H NMR) spectra, carbon nuclear magnetic resonance (^{13}C NMR) spectra, and fluorous nuclear magnetic resonance (^{19}F NMR) spectra were recorded at 400, 100, and 376 MHz, respectively. The ^{1}H NMR spectrum uses TMS (δ = 0.00 ppm) as an internal standard, and the ^{13}C NMR spectrum uses $CDCl_3$ (δ = 77.00 ppm) as an internal standard. Data are presented as follows: chemical shift (ppm), multiplicity (s = singlet, d = doublet, t = triplet, q = quartet, m = multiplet, br = broad, hept = heptet), coupling constants in Hertz (Hz), and integration. IR spectra were recorded on a PerkinElmer PE-983 spectrometer with absorption in cm^{-1}. High-resolution mass spectra (HRMS) were recorded by ESI and MALDI methods. The employed solvents were dried by standard methods when necessary. Commercially obtained reagents were used without further purification. Petroleum ether refers to the fraction with the boiling point in the range of 60−90 °C. For thin-layer chromatography (TLC), silica gel plates (Huanghai GF254) were used. Column chromatography was carried out using a 300−400 mesh silica gel at increased pressure.

The preparation of tropsulfimines **1**,[60] vinylidenecyclopropane-diester **2**,[65] tropones **4**,[66] and biscyano heptafulvene **1ae**[73] followed the previous literature procedure.

Compounds **1a**,[60] **1d**,[60] **1h**,[60] **1s**,[60] **1b**,[67] **1c**,[67] **1f**,[67] **1n**,[67] **4a**,[66] **4d**,[66] **4k**,[66] **4b**,[68] **4g**,[68] **4h**,[69] **4l**,[70] **4m**,[71] and **4q**[72] have been previously reported. Compounds **4n** and **4r** are commercially available.

*General Procedure for the Synthesis of Tropsulfimines **1e**, **1g**, **1i−1m**, **1o−1r**, and **1t−1x** from Tropone and Sulfonamides.* **Tropone 4n** (0.19 mL, 2.0 mmol, 1.0 equiv) was added to a suspension of **the corresponding sulfonamide** (2.0 mmol, 1.0 equiv) in 1,2-dichloroethane (10 mL), and the mixture was cooled to 0 °C in an ice bath. Titanium(IV) chloride (0.24 mL, 2.2 mmol, 1.1 equiv) was added slowly at 0 °C, and triethylamine (0.61 mL, 4.4 mmol, 2.2 equiv) was then added dropwise. The resulting brown heterogeneous mixture was refluxed in an oil bath for 4 h. After cooling to room temperature, the reaction mixture was diluted with dichloromethane (20 mL) and washed with saturated Na_2CO_3 solution (20 mL) and filtered through a pad of Celite to remove insoluble solids. The organic phases were then separated and the aqueous layer was extracted with dichloromethane (3 × 20 mL). The combined organic layers were dried over anhydrous Na_2SO_4, filtered, and evaporated. The residue was then purified by flash chromatography on silica gel (petroleum ether/ethyl acetate = 3/1) to afford the corresponding compound **tropsulfimines 1e, 1g, 1i−1m, 1o−1r**, and **1t−1x** as brown-yellow to green-yellow solids in 18−73% yields.

*N-(Cyclohepta-2,4,6-trien-1-ylidene)-4-(trifluoromethoxy)benzenesulfonamide (**1e**).* Yield: 0.2429 g, 37%; a brown solid; mp: 115−118 °C; Eluent:PE/EA = 3:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 8.06 (d, J = 8.6 Hz, 2H), 7.66 (br, 2H), 7.32 (d, J = 8.6 Hz, 2H), 7.31−7.24 (m, 2H), 7.20−7.11 (m, 2H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$, TMS) δ 170.6, 151.6, 140.7, 139.5, 137.1, 128.7, 120.6, 120.1 (q, J = 257.9 Hz); ^{19}F NMR (376 MHz, $CDCl_3$) δ −57.7; IR (neat): ν 2923, 2849, 1628, 1592, 1523, 1496, 1478, 1400, 1250, 1203, 1136, 1081, 856, 768, 713, 678 cm^{-1}; HRMS (ESI-TOF) m/z: [M + H]$^+$ Calcd for $C_{14}H_{11}NO_3F_3S$ 330.0406; found 330.0400.

*4-Bromo-N-(cyclohepta-2,4,6-trien-1-ylidene)-benzenesulfonamide (**1g**).* Yield: 0.4040 g, 63%; a yellow solid; mp: 119−122 °C; Eluent:PE/EA = 2:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 7.87 (dt, J = 8.6, 2.2 Hz, 2H), 7.69 (br, 2H), 7.64 (dt, J = 8.6, 2.2 Hz, 2H), 7.32−7.21 (m, 2H), 7.19−7.10 (m, 2H); ^{13}C{^{1}H} NMR (00 MHz, $CDCl_3$, TMS) δ 170.6, 141.4, 139.4, 137.0, 131.9, 128.3, 127.0; IR (neat): ν 3040, 2962, 1623, 1574, 1523, 1498, 1477, 1400, 1290, 1255, 1137, 1077, 861, 744 cm^{-1}; HRMS (ESI-TOF) m/z: [M + H]$^+$ Calcd for $C_{13}H_{11}NO_2SBr$ 323.9688; found 323.9694.

*N-(Cyclohepta-2,4,6-trien-1-ylidene)-4-vinylbenzenesulfonamide (**1i**).* Yield: 0.1899 g, 18%; a yellow solid; mp: 107−111 °C; Eluent:PE/EA = 3:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 7.96 (d, J = 8.2 Hz, 2H), 7.70 (br, 2H), 7.51 (d, J = 8.2 Hz, 2H), 7.29−7.18 (m, 2H), 7.14−7.04 (m, 2H), 6.74 (dd, J = 17.6, 10.9 Hz, 1H), 5.85 (d, J = 17.6 Hz, 1H), 5.39 (d,

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J = 10.9 Hz, 1H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 170.5, 141.4, 141.1, 139.0, 136.8, 135.5, 127.0, 126.4, 116.9; IR (neat): ν 2919, 1626, 1528, 1495,1477, 1396, 1282, 1250, 1141, 1083, 842, 763 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + H]^+$ Calcd for $C_{15}H_{14}NO_2S$ 272.0740; found 272.0734.

*N-(Cyclohepta-2,4,6-trien-1-ylidene)-4-ethynylbenzenesulfonamide (**1j**).* Yield: 0.1971g, 18%; a yellow solid; mp: 116−120 °C; Eluent:PE/EA = 3:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 7.96 (d, J = 8.0 Hz, 2H), 7.74 (br, 2H), 7.60 (d, J = 8.0 Hz, 2H), 7.31−7.21 (m, 2H), 7.12 (t, J = 4.8 Hz, 2H), 3.22 (s, 1H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 170.6, 142.3, 139.3, 137.0, 132.3, 126.7, 126.2, 82.3, 80.1; IR (neat): ν 2972, 2106, 1629, 1524, 1495, 1475, 1393, 1292, 1271, 1251, 1139, 1078, 855, 770 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + H]^+$ Calcd for $C_{15}H_{12}NO_2S$ 270.0583; found 270.0577.

*N-(Cyclohepta-2,4,6-trien-1-ylidene)-3-fluorobenzenesulfonamide (**1k**).* Yield: 0.2026 g; 19%; a brown-yellow solid; mp: 111−114 °C; Eluent:PE/EA = 3:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 7.81 (d, J = 7.7 Hz, 1H), 7.72 (dt, J = 8.5, 2.1 Hz, 1H), 7.65 (br, 2H), 7.49 (td, J = 8.1, 5.3 Hz, 1H), 7.33−7.20 (m, 3H), 7.19−7.11 (m, 2H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 170.8, 162.1 (d, J = 250.3 Hz), 144.2 (d, J = 6.7 Hz), 139.5, 137.1, 130.4 (d, J = 7.7 Hz), 122.4 (d, J = 3.3 Hz), 119.3 (d, J = 21.2 Hz), 114.1 (d, J = 24.5 Hz); ^{19}F NMR (376 MHz, $CDCl_3$) δ −110.6; IR (neat): ν 3092, 3071, 1628, 1589, 1527, 1477, 1316, 1307, 1223, 1151, 1086, 866, 789, 702 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + H]^+$ Calcd for $C_{13}H_{11}NO_2FS$ 264.0489; found 264.0492.

*3-Bromo-N-(cyclohepta-2,4,6-trien-1-ylidene)-benzenesulfonamide (**1l**).* Yield: 0.2590 g, 20%; a green solid; mp: 120−121 °C; Eluent:PE/EA = 2:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 8.16 (t, J = 1.8 Hz, 1H), 7.95 (d, J = 7.7 Hz, 1H), 7.68 (d, J = 7.7 Hz, 2H), 7.64 (br, 2H), 7.38 (t, J = 8.0 Hz, 1H), 7.34−7.24 (m, 2H), 7.20−7.12 (m, 2H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 170.7, 144.2, 139.4, 137.0, 135.2, 130.2, 129.8, 125.3, 122.6; IR (neat): ν 2988, 2899, 1628, 1521, 1498, 1477, 1401, 1281, 1255, 1141, 1078, 865, 759 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + H]^+$ Calcd for $C_{13}H_{11}NO_2SBr$ 323.9688; found 323.9691.

*3-Cyano-N-(cyclohepta-2,4,6-trien-1-ylidene)-benzenesulfonamide (**1m**).* Yield: 0.5837 g, 54%; a yellow solid; mp: 130−131 °C; Eluent:PE/EA = 3:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 8.31 (t, J = 1.7 Hz, 1H), 8.24 (dt, J = 8.0, 1.5 Hz, 1H), 7.82 (dt, J = 7.8, 1.4 Hz, 1H), 7.71 (br, 2H), 7.64 (t, J = 7.9 Hz, 1H), 7.40−7.27 (m, 2H), 7.25−7.14 (m, 2H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 170.9, 144.1, 139.8, 137.3, 135.2, 130.7, 130.6, 129.7, 117.5, 113.2; IR (neat): ν 3069, 2983, 2231, 1626, 1523, 1497, 1476, 1398, 1285, 1255, 1132, 1083, 885, 870, 766 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + H]^+$ Calcd for $C_{14}H_{11}N_2O_2S$ 271.0536; found 271.0541.

*2-Bromo-N-(cyclohepta-2,4,6-trien-1-ylidene)-benzenesulfonamide (**1o**).* Yield: 0.2604 g, 20%; a yellow solid; mp: 147−150 °C; Eluent:PE/EA = 2.5:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 8.25 (d, J = 7.1 Hz, 1H), 7.72 (d, J = 7.1 Hz, 1H), 7.60 (br, 2H), 7.48−7.43 (m, 1H), 7.40−7.34 (m, 1H), 7.29−7.25 (m, 2H), 7.17−7.12 (m, 2H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 171.0, 141.2, 139.2, 137.0, 135.1, 133.1, 129.6, 127.3, 120.9. IR (neat): ν 3079, 2983, 1628, 1523, 1498, 1473, 1403, 1295, 1277, 1254, 1146, 1122, 1097, 1027, 858, 763 cm^{-1}; HRMS (EI-TOF) m/z: $[M]^+$ Calcd for $C_{13}H_{10}NO_2SBr$ 322.9610; found 322.9613.

*N-(Cyclohepta-2,4,6-trien-1-ylidene)thiophene-2-sulfonamide (**1p**).* Yield: 0.2142 g, 21%; a brown-yellow solid; mp: 119−124 °C; Eluent:PE/EA = 3:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 7.72 (d, J = 3.8 Hz, 1H), 7.66 (br, 2H), 7.55 (d, J = 5.0 Hz, 1H), 7.32−7.28 (m, 2H), 7.20−7.11 (m, 2H), 7.06 (t, J = 4.4 Hz, 1H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 170.5, 143.5, 139.5, 137.1, 131.3, 131.2, 126.9; IR (neat): ν 3110, 2917, 2849, 1623, 1508, 1493, 1474, 1391, 1288, 1253, 1129, 1011, 849, 738 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + H]^+$ Calcd for $C_{11}H_{10}NO_2S_2$ 252.0148; found 252.0153.

*N-(Cyclohepta-2,4,6-trien-1-ylidene)naphthalene-2-sulfonamide (**1q**).* Yield: 0.1822 g, 15.44%; a brown-yellow solid; mp: 167−170 °C; Eluent:PE/EA = 3:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 8.57 (s, 1H), 8.04−7.92 (m, 3H), 7.89 (d, J = 7.9 Hz, 1H), 7.75 (br, 2H), 7.65−7.53 (m, 2H), 7.24 (ddd, J = 15.7, 7.5, 3.7 Hz, 2H), 7.13−7.04 (m, 2H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 170.5, 139.1, 136.8, 134.7, 132.1, 129.3, 129.0, 128.4, 127.8, 127.4, 127.2, 122.7; IR (neat): ν 2969, 1628, 1529, 1500, 1478, 1408, 1274, 1258, 1113, 1069, 874, 747 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + H]^+$ Calcd for $C_{17}H_{14}NO_2S$ 296.0740; found 296.0733.

*N-(Cyclohepta-2,4,6-trien-1-ylidene)-2,4,6-trimethylbenzenesulfonamide (**1r**).* Yield: 0.2570 g, 44.7%; a dark-green solid; mp: 116−119 °C; Eluent:PE/EA = 3:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 8.00 (br, 2H), 7.22−7.09 (m, 2H), 7.08−6.99 (m, 2H), 6.93 (s, 2H), 2.67 (s, 6H), 2.28 (s, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 170.0, 141.5, 138.5, 138.4, 136.8, 136.3, 131.4, 22.7, 20.8; IR (neat): ν 2975, 2933, 1628, 1500, 1475, 1403, 1289, 1132, 1056, 851, 697 cm^{-1}; HRMS (ESI-FTMS) m/z: $[M + H]^+$ Calcd for $C_{16}H_{18}NO_2S$ 288.1053; found 288.1047.

*N-(Cyclohepta-2,4,6-trien-1-ylidene)ethanesulfonamide (**1t**).* Yield: 0.1600 g, 20%; an orange-green solid; mp: 69−71 °C; Eluent:PE/EA = 2:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 8.01 (s, 1H), 7.35−6.64 (m, 5H), 3.17 (q, J = 7.4 Hz, 2H), 1.45 (t, J = 7.4 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 170.3, 141.7, 138.5, 135.9 (t, J = 101 Hz), 49.1, 7.9; IR (neat): ν 3024, 2933, 1631, 1528, 1502, 1475, 1406, 1256, 1106, 864, 763 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + H]^+$ Calcd for $C_9H_{12}NO_2S$ 198.0583; found 198.0578.

*N-(Cyclohepta-2,4,6-trien-1-ylidene)propane-2-sulfonamide (**1u**).* Yield: 0.2973 g, 33%; a yellow solid; mp: 69−73 °C; Eluent:PE/EA = 3:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 8.04 (s, 1H), 7.21−7.13 (m, 2H), 7.10 (br, 1H), 7.09−7.02 (m, 2H), 3.26 (hept, J = 6.6 Hz, 1H), 1.45 (d, J = 6.6 Hz, 6H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 170.9, 143.8, 138.3, 133.0, 54.5, 16.3; IR (neat): ν 2980, 2938, 2870, 1628, 1528, 1503, 1477, 1404, 1320, 1255, 1138, 1103, 857, 770, 711 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{10}H_{13}NO_2NaS$ 234.0559; found 234.0553.

*N-(Cyclohepta-2,4,6-trien-1-ylidene)-cyclopropanesulfonamide (**1v**).* Yield: 0.3058 g, 73%; a green-yellow solid; mp: 87−89 °C; Eluent:PE/EA = 3.5:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 8.00 (br, 1H), 7.28−6.96 (m, 5H), 2.69 (td, J = 8.0, 4.2 Hz, 1H), 1.24 (dt, J = 6.6, 4.2 Hz, 2H), 1.04 (dt, J = 8.0, 4.4 Hz, 2H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 170.2, 142.4, 138.5, 135.7, 32.1, 5.3; IR (neat): ν 3048, 3011, 2962, 1626, 1524, 1500, 1475, 1409, 1262, 1115, 1046, 858, 770, 676 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + H]^+$ Calcd for $C_{10}H_{12}NO_2S$ 210.0583; found 210.0585.

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*N-(Cyclohepta-2,4,6-trien-1-ylidene)-2-methylpropane-2-sulfonamide (**1w**).* Yield: 0.3893 g, 43%; a yellow solid; mp: 129−132 °C; Eluent:PE/EA = 3:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 8.05 (d, J = 12.1 Hz, 1H), 7.15−6.98 (m, 5H), 1.49 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 171.1, 142.2, 138.1, 137.9, 136.7, 135.6, 135.3, 58.5, 24.0; IR (neat): ν 2985, 2923, 1634, 1526, 1500, 1476, 1407, 1253, 1229, 1200, 1092, 856, 773 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{11}H_{15}NO_2NaS$ 248.0716; found 248.0717.

*N-(Cyclohepta-2,4,6-trien-1-ylidene)-1,1,1-trifluoromethanesulfonamide (**1x**).* Yield: 0.4131 g, 44%; a red solid; mp: 105−108 °C; Eluent:PE/EA = 3:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 7.78 (d, J = 11.5 Hz, 2H), 7.61−7.53 (m, 2H), 7.48−7.41 (m, 2H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 172.9, 142.0, 140.0, 139.1, 119.1 (q, J = 319.1 Hz); ^{19}F NMR (376 MHz, $CDCl_3$) δ −79.3; IR (neat): ν 2920, 1621, 1501, 1480, 1377, 1321, 1177, 1118, 891, 877, 768 cm^{-1}; HRMS (ESI-FTMS) m/z: $[M + H]^+$ Calcd for $C_8H_6NO_2SF_3$ 237.0066; found 237.0063.

*General Procedure for the Preparation of Tropones **4c**, **4e**, **4f**, **4i**, and **4j**.* **2-Chloro-2,4,6-cycloheptatrien-1-one 4r** (0.70 g, 5.0 mmol, 1.0 equiv), **the corresponding arylboronic acid** (7.5 mmol, 1.5 equiv), and Cs_2CO_3 (1.63 g, 5.0 mmol, 1.0 equiv) were added to a solution of $Pd(PPh_3)_4$ (0.28 g, 0.25 mmol, 5 mol %) in anhydrous THF (25 mL) under an argon atmosphere. The resulting solution was stirred for 4 h at 60 °C in an oil bath. After cooling to room temperature, the solution was filtered through a pad of Celite to remove insoluble solids. The filtrate was concentrated under reduced pressure, and the residue was then purified by flash chromatography on silica gel (petroleum ether/ethyl acetate = 3/1) to give **4c**, **4e**, **4f**, **4i**, and **4j** as pale-yellow solids in 74−83% yield.

*General Procedure for the Preparation of Tropone **4s**.* **Hinokitiol** (3.28 g, 20.0 mmol, 1.0 equiv) was dissolved in anhydrous toluene (120 mL). $SOCl_2$ (2.2 mL, 30 mmol, 1.5 equiv) was slowly added, and the mixture was stirred for 4 h at 90 °C in an oil bath. The solvent was removed under vacuum, and the crude product was purified by flash chromatography to give **2-chloro-4-(2-chloropropan-2-yl)cyclohepta-2,4,6-trien-1-one (S-1)** as a brown oil in 39% yield (1.6934 g). **S-1** (1.08 g, 5.0 mmol, 1.0 equiv) and Cs_2CO_3 (1.63 g, 5.0 mmol, 1.0 equiv) were added to a solution of $Pd(PPh_3)_4$ (0.28 g, 0.25 mmol, 5 mol %) in anhydrous THF (25 mL) under an argon atmosphere. The resulting solution was stirred for 4 h at 60 °C in an oil bath. After cooling to room temperature, the solution was filtered through a pad of Celite to remove insoluble solids. The filtrate was concentrated under reduced pressure, and the residue was then purified by flash chromatography on silica gel (petroleum ether/ethyl acetate = 3/1) to give **2-chloro-4-(prop-1-en-2-yl)cyclohepta-2,4,6-trien-1-one 4s** as a brown oil in 38% yield (0.3470 g).

*General Procedure for the Preparation of Tropone **4t**.* **Hinokitiol** (1.64 g, 10.0 mmol, 1.0 equiv) was added to a solution of TsCl (1.90 g, 10.0 mmol, 1.0 equiv) in DCM (30 mL). TEA (1.4 mL, 10.0 mmol, 1.0 equiv) was slowly added to the solution, and the resulting solution was stirred for 12 h at room temperature. After usual workup and passing through flash chromatography on silica gel (petroleum ether/ethyl acetate = 4/1), **3-isopropyl-7-oxocyclohepta-1,3,5-trien-1-yl 4-methylbenzenesulfonate (S-2)** could be obtained as a white solid in 50% yield (1.5875 g). **S-2** (1.58 g, 5 mmol, 1.0 equiv) was dissolved in a solution of HCl in 1,4-dioxane (4.0 M, 10 mL) and heated to 80 °C in an oil bath for 2 h. After usual workup and passing through flash chromatography on silica gel (petroleum ether/ethyl acetate = 3/1), the final compound **2-chloro-4-isopropylcyclohepta-2,4,6-trien-1-one 4t** could be obtained as a yellow oil in 47% yield (0.4290 g).

*2-(4-Chlorophenyl)cyclohepta-2,4,6-trien-1-one (**4c**).* Yield: 0.8045 g, 74%; a pale-yellow solid; mp: 89−93 °C; Eluent:PE/EA = 3:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 7.43 (d, J = 8.8 Hz, 2H), 7.39−7.30 (m, 3H), 7.20−7.10 (m, 2H), 7.06−6.95 (m, 2H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 186.1, 151.1, 142.3, 138.2, 136.5, 135.3, 134.3, 133.6, 133.5, 130.5, 128.2; IR (neat): ν 1628, 1572, 1516, 1487, 1462, 1380, 1258, 1085, 1013, 780 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + H]^+$ Calcd for $C_{13}H_{10}OCl$ 217.0415; found 217.0414.

*4-(7-Oxocyclohepta-1,3,5-trien-1-yl)benzaldehyde (**4e**).* Yield: 0.8719 g, 83%; a pale-yellow solid; mp: 88−89 °C; Eluent:PE/EA = 3:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 10.04 (s, 1H), 7.91 (d, J = 8.0 Hz, 2H), 7.64 (d, J = 8.0 Hz, 2H), 7.37 (dd, J = 8.0, 1.8 Hz, 1H), 7.23−7.16 (m, 2H), 7.10−7.02 (m, 2H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 191.8, 185.9, 151.2, 146.1, 142.7, 136.9, 135.8, 135.5, 134.2, 133.5, 129.8, 129.4; IR (neat): ν 1691, 1628, 1603, 1574, 1461, 1380, 1212, 823 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + H]^+$ Calcd for $C_{14}H_{11}O_2$ 211.0754; found 211.0744.

*Methyl 4-(7-Oxocyclohepta-1,3,5-trien-1-yl)benzoate (**4f**).* Yield: 0.9879 g, 82%; a pale-yellow solid; mp: 89−92 °C; Eluent:PE/EA = 3:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 8.06 (d, J = 8.0 Hz, 2H), 7.55 (d, J = 8.0 Hz, 2H), 7.35 (d, J = 8.2 Hz, 1H), 7.16 (d, J = 6.6 Hz, 2H), 7.09−6.94 (m, 2H), 3.92 (s, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 186.0, 166.6, 151.3, 144.4, 142.4, 136.7, 135.4, 133.9, 133.5, 129.6, 129.2, 129.0, 52.0; IR (neat): ν 2956, 1709, 1631, 1576, 1519, 1383, 1273, 1193, 1108, 709 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + H]^+$ Calcd for $C_{15}H_{13}O_3$ 241.0859; found 241.0857.

*2-(7-Oxocyclohepta-1,3,5-trien-1-yl)benzonitrile (**4i**).* Yield: 0.8296 g, 80%; a pale-yellow solid; mp: 92−95 °C; Eluent:PE/EA = 3:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 7.77 (t, J = 2.0 Hz, 1H), 7.70 (dt, J = 7.8, 1.6 Hz, 1H), 7.63 (dt, J = 7.8, 1.6 Hz, 1H), 7.49 (t, J = 7.8 Hz, 1H), 7.34−7.30 (m, 1H), 7.20−7.16 (m, 2H), 7.08−7.02 (m, 2H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 185.7, 150.0, 142.8, 140.9, 137.1, 135.7, 134.4, 133.6, 133.5, 132.7, 131.7, 128.8, 118.6, 112.2; IR (neat): ν 2226, 1628, 1564, 1519, 1463, 1389, 1262, 802 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + H]^+$ Calcd for $C_{14}H_{10}NO$ 208.0757; found 208.0747.

*2-(2-Chlorophenyl)cyclohepta-2,4,6-trien-1-one (**4j**).* Yield: 0.8512 g, 79%; a pale-yellow solid; mp: 87−90 °C; Eluent:PE/EA = 3:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 7.45−7.38 (m, 1H), 7.31−7.22 (m, 4H), 7.18−7.12 (m, 2H), 7.03−6.97 (m, 2H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 185.7, 151.2, 142.0, 139.2, 137.0, 135.4, 134.2, 133.4, 132.1, 130.7, 129.3, 129.1, 126.7; IR (neat): ν 1626, 1572, 1521, 1460, 1386, 1033, 768 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + H]^+$ Calcd for $C_{13}H_{10}OCl$ 217.0415; found 217.0408.

*2-Chloro-4-(prop-1-en-2-yl)cyclohepta-2,4,6-trien-1-one (**4s**).* Yield: 0.3470 g, 38%; a brown oil; Eluent:PE/EA = 3:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 8.03 (s, 1H), 7.26−7.19 (m, 1H), 7.17−7.07 (m, 2H), 5.38 (s, 1H), 5.31 (s, 1H), 2.13 (s, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 179.8, 148.0, 145.7, 144.1, 137.0, 135.8, 135.5, 129.9, 117.8, 22.0; IR (neat): ν 2980, 2930, 1631, 1590, 1514, 1461, 1383, 1221, 897 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + H]^+$ Calcd for $C_{10}H_{10}OCl$ 181.0415; found 181.0408.

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*2-Chloro-4-isopropylcyclohepta-2,4,6-trien-1-one (**4t**).* Yield: 0.4290 g, 47%; a yellow oil; Eluent:PE/EA = 3:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 7.77 (t, J = 1.6 Hz, 1H), 7.28−7.03 (m, 2H), 6.95 (d, J = 8.2 Hz, 1H), 2.82 (hept, J = 7.0 Hz, 1H), 1.26 (dt, J = 7.0, 1.4 Hz, 6H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$, TMS) δ 179.7, 152.9, 148.6, 137.2, 136.5, 136.3, 130.1, 38.1, 22.8; IR (neat): ν 2962, 2930, 2870, 1627, 1588, 1501, 1464, 1388, 1268, 896 cm^{-1}; HRMS (ESI-TOF) m/z: [M + H]$^+$ Calcd for $C_{10}H_{12}OCl$ 183.0571; found 183.0571.

*General Procedure for the Synthesis of Compounds **3**.* Tris(dibenzylideneacetone)dipalladium (0.0023 g, 0.002 mmol, 2.5 mol %) and DPEPhos (0.0040 g, 0.007 mmol, 7.5 mol %) were added to an oven-dried 10 mL sealed tube equipped with a magnetic stirring bar under an argon atmosphere. The resulting catalyst system was stirred in THF (degassed) (1.0 mL) for 30 min. After that, **1** (0.1 mmol), **2** (0.2 mmol), $Yb(OTf)_3$ (0.0062 g, 0.01 mmol, 10 mol %), and Cs_2CO_3 (0.0325 g, 0.1 mmol, 1.0 equiv) were added under an argon atmosphere. The reaction mixture was stirred at 70 °C in an oil bath for 12 h. The reaction solution was filtered through a pad of Celite to remove insoluble solids. Then, the solvent was evaporated under vacuum, and the residue was purified by column chromatography on silica gel (petroleum ether/ethyl acetate = 3/1) to afford the desired product **3** as a white solid.

*1-Tosyl-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diethyl (**3a**).* Yield: 0.0431 g, 91%; a colorless solid; mp: 123−127 °C; Eluent:PE/EA = 4:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 7.27 (d, J = 8.0 Hz, 2H), 7.15 (d, J = 8.0 Hz, 2H), 6.84 (dd, J = 10.7, 6.0 Hz, 1H), 6.71 (d, J = 6.5 Hz, 1H), 6.63 (dd, J = 10.8, 5.8 Hz, 1H), 6.08 (dd, J = 8.8, 5.8 Hz, 1H), 5.28 (dd, J = 11.5, 4.2 Hz, 1H), 5.14 (dd, J = 11.5, 4.0 Hz, 1H), 4.23−4.06 (m, 5H), 2.75 (d, J = 13.9 Hz, 1H), 2.39 (s, 3H), 2.29 (dt, J = 13.8, 4.2 Hz, 1H), 1.71 (d, J = 7.6 Hz, 1H), 1.22 (t, J = 7.1 Hz, 3H), 1.12 (t, J = 7.1 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$, TMS) δ 207.6, 169.3, 168.8, 144.1, 134.2, 129.4, 129.1, 128.2, 127.2, 126.5, 117.0, 114.3, 113.9, 102.1, 81.2, 61.9, 61.7, 55.9, 40.4, 30.5, 21.6, 14.1, 14.0; IR (neat): ν 2983, 2923, 1968, 1728, 1587, 1493, 1443, 1365, 1253, 1164, 1133, 1090, 872, 713, 657 cm^{-1}; HRMS (ESI-TOF) m/z: [M + Na]$^+$ Calcd for $C_{25}H_{27}NO_6NaS$ 492.1451; found 492.1444.

*1-(Phenylsulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diethyl (**3b**).* Yield: 0.0366 g, 80%; a colorless solid; mp: 117−119 °C; Eluent:PE/EA = 4:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 7.52 (d, J = 7.0 Hz, 1H), 7.41−7.33 (m, 4H), 6.84 (dd, J = 10.8, 6.4 Hz, 1H), 6.74 (d, J = 6.4 Hz, 1H), 6.64 (dd, J = 10.8, 5.8 Hz, 1H), 6.07 (dd, J = 8.8, 5.8 Hz, 1H), 5.30 (dd, J = 11.6, 4.4 Hz, 1H), 5.16 (dd, J = 11.6, 3.8 Hz, 1H), 4.28−4.03 (m, 5H), 2.76 (d, J = 14.0 Hz, 1H), 2.28 (dt, J = 14.0, 4.2 Hz, 1H), 1.71 (d, J = 7.7 Hz, 1H), 1.22 (t, J = 7.1 Hz, 3H), 1.11 (t, J = 7.1 Hz, 3H); ^{13}C{^{1}H} NMR (400 MHz, $CDCl_3$, TMS) δ 207.6, 169.3, 168.7, 133.2, 129.4, 128.5, 128.1, 127.3, 126.5, 116.9, 114.3, 113.9, 102.0, 81.3, 61.9, 61.7, 55.8, 40.3, 30.5, 14.1, 13.9; IR (neat): ν 2988, 2899, 2009, 1733, 1592, 1446, 1407, 1301, 1147, 1088, 961, 769 cm^{-1}; HRMS (ESI-TOF) m/z: [M + Na]$^+$ Calcd for $C_{24}H_{25}NO_6NaS$ 478.1295; found 478.1293.

*1-((4-(tert-Butyl)phenyl)sulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diethyl (**3c**).* Yield: 0.0415 g, 81%; a colorless solid; mp: 121−123 °C; Eluent:PE/EA = 4:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 7.36 (d, J = 8.8 Hz, 2H), 7.32 (d, J = 8.8 Hz, 2H), 6.82 (dd, J = 10.8, 6.4 Hz, 1H), 6.72 (d, J = 6.4 Hz, 1H), 6.59 (dd, J = 10.8, 5.8 Hz, 1H), 5.97 (dd, J = 8.8, 5.8 Hz, 1H), 5.29 (dd, J = 11.4, 4.2 Hz, 1H), 5.15 (dd, J = 11.4, 3.8 Hz, 1H), 4.35−4.00 (m, 5H), 2.79 (dd, J = 13.9, 1.7 Hz, 1H), 2.39 (dt, J = 13.9, 4.1 Hz, 1H), 1.70 (d, J = 7.5 Hz, 1H), 1.32 (s, 9H), 1.22 (t, J = 7.1 Hz, 3H), 1.11 (t, J = 7.1 Hz, 3H); ^{13}C{^{1}H} NMR (400 MHz, $CDCl_3$, TMS) δ 207.6, 169.3, 168.8, 157.1, 134.1, 129.4, 128.0, 127.1, 126.4, 125.4, 117.2, 114.1, 113.5, 102.1, 81.3, 61.9, 61.7, 55.9, 40.4, 35.2, 31.0, 30.6, 14.1, 14.0; IR (neat): ν 3027, 2964, 2025, 1733, 1594, 1493, 1440, 1362, 1306, 1149, 1088, 955, 759 cm^{-1}; HRMS (ESI-TOF) m/z: [M + Na]$^+$ Calcd for $C_{28}H_{33}NO_6NaS$ 534.1921; found 534.1928.

*1-((4-Methoxyphenyl)sulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diethyl (**3d**).* Yield: 0.0421 g, 87%; a colorless solid; mp: 119−120 °C; Eluent:PE/EA = 4:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 7.32 (d, J = 8.5 Hz, 2H), 6.82 (m, 3H), 6.72 (d, J = 6.6 Hz, 1H), 6.64 (dd, J = 10.8, 5.8 Hz, 1H), 6.10 (dd, J = 8.8, 5.8 Hz, 1H), 5.29 (dd, J = 11.5, 4.4 Hz, 1H), 5.15 (dd, J = 11.5, 3.8 Hz, 1H), 4.22−4.06 (m, 5H), 3.85 (s, 3H), 2.77 (d, J = 14.0 Hz, 1H), 2.32 (dt, J = 14.0, 4.2 Hz, 1H), 1.71 (d, J = 7.7 Hz, 1H), 1.22 (t, J = 7.1 Hz, 3H), 1.12 (t, J = 7.1 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$, TMS) δ 207.6, 169.3, 168.8, 163.3, 130.2, 129.4, 128.8, 127.1, 126.4, 117.1, 114.4, 113.8, 113.6, 102.1, 81.1, 61.9, 61.7, 55.8, 55.6, 40.4, 30.5, 14.1, 14.0; IR (neat): ν 2983, 2933, 2155, 1732, 1593, 1493, 1440, 1259, 1161, 1090, 801, 717 cm^{-1}; HRMS (ESI-TOF) m/z: [M + Na]$^+$ Calcd for $C_{25}H_{27}NO_7NaS$ 508.1400; found 508.1400.

*1-((4-(Trifluoromethoxy)phenyl)sulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diethyl (**3e**).* Yield: 0.0485 g, 90%; a colorless solid; mp: 129−132 °C; Eluent:PE/EA = 4:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 7.45 (d, J = 8.5 Hz, 2H), 7.18 (d, J = 8.5 Hz, 2H), 6.83 (dd, J = 10.7, 6.4 Hz, 1H), 6.74 (d, J = 6.5 Hz, 1H), 6.63 (dd, J = 10.8, 5.8 Hz, 1H), 6.02 (dd, J = 8.7, 5.9 Hz, 1H), 5.31 (dd, J = 11.6, 4.2 Hz, 1H), 5.18 (dd, J = 11.6, 3.8 Hz, 1H), 4.23−4.05 (m, 5H), 2.83 (d, J = 14.0 Hz, 1H), 2.39 (dt, J = 14.0, 4.2 Hz, 1H), 1.74 (d, J = 7.6 Hz, 1H), 1.23 (t, J = 7.6 Hz, 3H), 1.12 (t, J = 7.2 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$, TMS) δ 207.5, 169.2, 168.6, 152.5 (q, J = 2.1 Hz), 135.3, 130.2, 129.4, 127.5, 126.6, 120.3, 120.1 (q, J = 258.0 Hz), 116.9, 114.2, 113.7, 101.9, 81.6, 62.0, 61.8, 55.8, 40.3, 30.6, 14.1, 13.9; ^{19}F NMR (376 MHz, $CDCl_3$) δ −57.8; IR (neat): ν 2977, 2920, 2012, 1734, 1589, 1474, 1433, 1369, 1259, 1225, 1168, 1066, 788, 697 cm^{-1}; HRMS (ESI-TOF) m/z: [M + Na]$^+$ Calcd for $C_{25}H_{24}NO_7F_3NaS$ 562.1118; found 562.1110.

*1-((4-Chlorophenyl)sulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diethyl (**3f**).* Yield: 0.0381g, 78%; a colorless solid; mp: 124−126 °C; Eluent:PE/EA = 4:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 7.38−7.28 (m, 4H), 6.84 (dd, J = 10.8, 6.4 Hz, 1H), 6.72 (d, J = 6.5 Hz, 1H), 6.65 (dd, J = 10.8, 5.8 Hz, 2H), 6.09 (dd, J = 8.8, 5.8 Hz, 1H), 5.30 (dd, J = 11.6, 4.2 Hz, 1H), 5.17 (dd, J = 11.6, 4.0 Hz, 1H), 4.26−4.07 (m, 5H), 2.80 (dd, J = 13.9, 1.6 Hz, 1H), 2.34 (dt, J = 13.9, 4.2 Hz, 1H), 1.73 (d, J = 7.2 Hz, 1H), 1.23 (t, J = 7.1 Hz, 3H), 1.13 (t, J = 7.1 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$, TMS) δ 207.5, 169.2, 168.7, 139.8, 135.5, 129.5, 128.8, 127.5, 126.7, 116.8, 114.3,

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113.9, 101.9, 81.5, 62.0, 61.8, 55.8, 40.4, 30.6, 14.1, 14.0; IR (neat): ν 2982, 2019, 1733, 1584, 1474, 1438, 1367, 1180, 1167, 1092, 756, 713 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{24}H_{24}NO_6NaSCl$ 512.0905; found 512.0905.

*1-((4-Bromophenyl)sulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diethyl (**3g**).* Yield: 0.0468 g, 88%; a colorless solid; mp: 136−139 °C; Eluent:PE/EA = 4:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 7.50 (d, J = 8.6 Hz, 2H), 7.25 (d, J = 8.5 Hz, 2H), 6.84 (dd, J = 10.8, 6.4 Hz, 1H), 6.72 (d, J = 6.5 Hz, 1H), 6.65 (dd, J = 10.8, 5.8 Hz, 1H), 6.09 (dd, J = 8.6, 5.9 Hz, 1H), 5.30 (dd, J = 11.6, 4.4 Hz, 1H), 5.17 (dd, J = 11.6, 4.0 Hz, 1H), 4.31−4.07 (m, 5H), 2.81 (d, J = 14.0 Hz, 1H), 2.35 (dt, J = 14.0, 4.1 Hz, 1H), 1.73 (d, J = 7.6 Hz, 1H), 1.23 (t, J = 7.1 Hz, 3H), 1.13 (t, J = 7.1 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 207.5, 169.2, 168.7, 136.0, 131.8, 129.5, 129.4, 128.3, 127.5, 126.7, 116.8, 114.3, 113.8, 101.9, 81.6, 62.0, 61.8, 55.8, 40.4, 30.6, 14.1, 14.0; IR (neat): ν 2962, 2925, 2024, 1733, 1570, 1469, 1443, 1366, 1259, 1166, 1086, 1012, 798,743 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{24}H_{24}NO_6NaSBr$ 556.0400; found 556.0399.

*1-((4-Nitrophenyl)sulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diethyl (**3h**).* Yield: 0.0315 g, 63%; a colorless solid; mp: 157−161 °C; Eluent:PE/EA = 4:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 8.21 (d, J = 8.8 Hz, 2H), 7.58 (d, J = 8.8 Hz, 2H), 6.87 (dd, J = 10.8, 6.4 Hz, 1H), 6.76 (d, J = 6.4 Hz, 1H), 6.68 (dd, J = 10.8, 5.8 Hz, 1H), 6.08 (dd, J = 8.8, 5.8 Hz, 1H), 5.34 (dd, J = 11.8, 4.2 Hz, 1H), 5.21 (dd, J = 11.8, 3.8 Hz, 1H), 4.29−4.03 (m, 5H), 2.82 (dd, J = 14.0, 1.6 Hz, 1H), 2.29 (dt, J = 14.0, 4.2 Hz, 1H), 1.76 (d, J = 7.5 Hz, 1H), 1.23 (t, J = 7.0 Hz, 3H), 1.12 (t, J = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 207.4, 169.1, 168.5, 150.2, 142.6, 129.6, 129.3, 128.0, 127.0, 123.6, 116.5, 114.5, 114.2, 101.7, 82.0, 62.1, 62.0, 55.7, 40.4, 30.7, 14.1, 14.0; IR (neat): ν 2980, 2061, 1734, 1532, 1440, 1367, 1347, 1260, 1179, 1036, 851, 736 cm^{-1}; HRMS (ESI-FTMS) m/z: $[M + H]^+$ $C_{24}H_{25}N_2O_8S$ Calcd for 501.1326; found 501.1319.

*2-(1λ5-Vinylidene)-1-((4-vinylphenyl)sulfonyl)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diethyl (**3i**).* Yield: 0.0405 g, 84%; a colorless solid; mp: 118−121 °C; Eluent:PE/EA = 4:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 7.36 (d, J = 8.6 Hz, 2H), 7.35 (d, J = 8.6 Hz, 2H), 6.84 (dd, J = 10.8, 6.4 Hz, 1H), 6.76−6.59 (m, 3H), 6.05 (dd, J = 8.8, 5.8 Hz, 1H), 5.86 (d, J = 17.6 Hz, 1H), 5.43 (d, J = 11.0 Hz, 1H), 5.30 (dd, J = 11.5, 4.2 Hz, 1H), 5.16 (dd, J = 11.5, 3.8 Hz, 1H), 4.26−4.05 (m, 5H), 2.78 (dd, J = 13.9, 1.6 Hz, 1H), 2.35 (dt, J = 13.9, 4.1 Hz, 1H), 1.72 (d, J = 7.5 Hz, 1H), 1.22 (t, J = 7.1 Hz, 3H), 1.11 (t, J = 7.1 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 207.6, 169.3, 168.8, 142.2, 135.8, 135.2, 129.4, 128.4, 127.3, 126.5, 126.0, 117.7, 116.9, 114.3, 113.8, 102.1, 81.3, 61.9, 61.7, 55.8, 40.4, 30.5, 14.1, 14.0; IR (neat): ν 2983, 2936, 1970, 1733, 1589, 1443, 1365, 1259, 1165, 1088, 715 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{26}H_{27}NO_6NaS$ 504.1451; found 504.1448.

*1-((4-Ethynylphenyl)sulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diethyl (**3j**).* Yield: 0.0394 g, 82%; a colorless solid; mp: 126−127 °C; Eluent:PE/EA = 4:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 7.45 (d, J = 8.1 Hz, 2H), 7.34 (d, J = 8.1 Hz, 2H), 6.84 (dd, J = 10.8, 6.4 Hz, 1H), 6.73 (d, J = 6.4 Hz, 1H), 6.64 (dd, J = 10.8, 5.8 Hz, 1H), 6.08 (dd, J = 8.8, 5.8 Hz, 1H), 5.30 (dd, J = 11.6, 4.2 Hz, 1H), 5.17 (dd, J = 11.6, 3.8 Hz, 1H), 4.27−4.03 (m, 5H), 3.27 (s, 1H), 2.79 (dd, J = 14.0, 1.6 Hz, 1H), 2.33 (dt, J = 14.0, 4.2 Hz, 1H), 1.73 (d, J = 7.2 Hz, 1H), 1.22 (t, J = 7.1 Hz, 3H), 1.12 (t, J = 7.1 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 207.5, 169.2, 168.7, 136.9, 132.0, 129.4, 128.0, 127.5, 127.2, 126.7, 116.8, 114.3, 113.9, 101.9, 81.9, 81.5, 81.2, 62.0, 61.8, 55.8, 40.4, 30.6, 14.1, 14.0; IR (neat): ν 2985, 2372, 1962, 1733, 1587, 1440, 1366, 1259, 1165, 1083, 718 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{26}H_{25}NO_6NaS$ 502.1295; found 502.1299.

*1-((3-Fluorophenyl)sulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diethyl (**3k**).* Yield: 0.0224 g, 47%; a colorless solid; mp: 125−126 °C; Eluent:PE/EA = 4:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 7.39 (td, J = 8.1, 5.2 Hz, 1H), 7.30−7.19 (m, 2H), 6.99 (dt, J = 8.1, 2.1 Hz, 1H), 6.86 (dd, J = 10.8, 6.4 Hz, 1H), 6.75 (d, J = 6.5 Hz, 1H), 6.68 (dd, J = 10.8, 5.8 Hz, 1H), 6.15 (dd, J = 8.8, 5.7 Hz, 1H), 5.31 (dd, J = 11.7, 4.4 Hz, 1H), 5.18 (dd, J = 11.7, 3.8 Hz, 1H), 4.26−4.06 (m, 5H), 2.78 (dd, J = 13.9, 1.6 Hz, 1H), 2.28 (dt, J = 13.9, 4.1 Hz, 1H), 1.74 (d, J = 8.0 Hz, 1H), 1.23 (t, J = 7.1 Hz, 3H), 1.12 (t, J = 7.1 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 207.6, 169.2, 168.7, 161.9 (d, J = 251.8 Hz), 138.7 (d, J = 6.9 Hz), 130.3 (d, J = 7.7 Hz), 129.5, 127.7, 126.6, 123.6 (d, J = 3.5 Hz), 120.3 (d, J = 21.1 Hz), 116.6, 115.9 (d, J = 24.6 Hz), 114.5, 114.2, 101.9, 81.6, 62.0, 61.8, 55.8, 40.3, 30.5, 14.1, 14.0; ^{19}F NMR (376 MHz, $CDCl_3$) δ −110.6; IR (neat): ν 2970, 2924, 1968, 1733, 1587, 1474, 1369, 1259, 1225, 1167, 1066, 872, 796, 700 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{24}H_{24}NO_6FNaS$ 496.1201; found 496.1199.

*1-((3-Bromophenyl)sulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diethyl (**3l**).* Yield: 0.0422 g, 84%; a colorless solid; mp: 138−140 °C; Eluent:PE/EA = 4:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 7.66 (dt, J = 8.0, 1.3 Hz, 1H), 7.47 (dt, J = 8.0, 1.3 Hz, 1H), 7.34−7.26 (m, 2H), 6.87 (dd, J = 10.8, 6.4 Hz, 1H), 6.75 (d, J = 6.4 Hz, 1H), 6.70 (dd, J = 10.8, 5.8 Hz, 1H), 6.23 (dd, J = 8.8, 5.8 Hz, 1H), 5.31 (dd, J = 11.7, 4.4 Hz, 1H), 5.17 (dd, J = 11.7, 3.8 Hz, 1H), 4.26−4.07 (m, 5H), 2.78 (dd, J = 14.0, 1.7 Hz, 1H), 2.24 (dt, J = 14.0, 4.1 Hz, 1H), 1.74 (dt, J = 7.5, 1.4 Hz, 1H), 1.22 (t, J = 7.1 Hz, 3H), 1.12 (t, J = 7.1 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 207.6, 169.2, 168.7, 138.5, 136.1, 131.4, 130.1, 129.5, 127.8, 126.6, 126.1, 122.5, 116.5, 114.6, 114.3, 101.9, 81.6, 62.0, 61.8, 55.8, 40.3, 30.4, 14.1, 14.0; IR (neat): ν 2988, 2901, 2022, 1733, 1592, 1501, 1443, 1369, 1259, 1179, 1066, 715 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{24}H_{24}NO_6NaSBr$ 556.0400; found 556.0406.

*1-((3-Cyanophenyl)sulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diethyl (**3m**).* Yield: 0.0423 g, 88%; a colorless solid; mp: 140−141 °C; Eluent:PE/EA = 4:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 7.81 (dt, J = 7.7, 1.4 Hz, 1H), 7.73 (dt, J = 8.0, 1.4 Hz, 1H), 7.57 (t, J = 8.0 Hz, 1H), 7.49 (d, J = 1.7 Hz, 1H), 6.89 (dd, J = 10.8, 6.4 Hz, 1H), 6.77 (d, J = 6.4 Hz, 1H), 6.72 (dd, J = 10.8, 5.8 Hz, 1H), 6.19 (dd, J = 8.8, 5.8 Hz, 1H), 5.33 (dd, J = 11.8, 4.4 Hz, 1H), 5.20 (dd, J = 11.8, 3.8 Hz, 1H), 4.33−3.98 (m, 5H), 2.80 (dd, J = 14.0, 1.6 Hz, 1H), 2.21 (dt, J = 14.0, 4.0 Hz, 1H), 1.75 (d, J = 7.5 Hz, 1H), 1.23 (t, J = 7.1 Hz, 3H), 1.12 (t, J = 7.1 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 207.5, 169.0, 168.5, 138.4, 136.1, 131.9, 131.4, 129.7, 129.6, 128.0, 126.8, 116.8, 116.3, 114.6, 114.3, 113.0, 101.6, 81.9, 62.1, 61.9, 55.7, 40.3, 30.6, 14.1, 14.0; IR (neat): ν 2985, 2899, 2234, 2020, 1733, 1592, 1469, 1369, 1260, 1187,

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1028, 890, 718 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{25}H_{24}N_2O_6NaS$ 503.1247; found 503.1246.

*1-(o-Tolylsulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diethyl (**3n**).* Yield: 0.0347 g, 74%; a colorless solid; mp: 124−125 °C; Eluent:PE/EA = 4:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 7.56 (d, J = 7.6 Hz, 1H), 7.40 (t, J = 7.4 Hz, 1H), 7.24−7.15 (m, 2H), 6.80 (dd, J = 10.8, 6.4 Hz, 1H), 6.65 (dd, J = 10.8, 5.8 Hz, 1H), 6.58 (d, J = 6.4 Hz, 1H), 6.20 (dd, J = 8.8, 5.7 Hz, 1H), 5.01 (dd, J = 11.5, 3.8 Hz, 1H), 4.95 (dd, J = 11.5, 4.2 Hz, 1H), 4.48 (t, J = 8.2 Hz, 1H), 4.26−4.08 (m, 4H), 2.84 (dd, J = 13.9, 1.6 Hz, 1H), 2.68 (dt, J = 13.9, 4.0 Hz, 1H), 2.45 (s, 3H), 1.79 (d, J = 7.8 Hz, 1H), 1.22 (t, J = 7.1 Hz, 3H), 1.15 (t, J = 7.1 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 207.1, 169.5, 168.8, 138.3, 135.7, 133.1, 132.4, 131.0, 129.2, 127.5, 126.6, 125.8, 116.6, 115.0, 114.3, 102.4, 81.0, 61.9, 61.8, 56.0, 40.6, 30.9, 20.9, 14.1, 14.0; IR (neat): ν 2980, 2928, 1960, 1733, 1589, 1440, 1363, 1258, 1180, 1165, 1028, 869, 759, 710 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{25}H_{27}NO_6NaS$ 492.1451; found 492.1456.

*1-(Thiophen-2-ylsulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diethyl (**3p**).* Yield: 0.0410 g, 89%; a colorless solid; mp: 127−130 °C; Eluent:PE/EA = 4:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 7.53 (dd, J = 5.0, 1.4 Hz, 1H), 7.06 (dd, J = 3.8, 1.4 Hz, 1H), 6.95 (dd, J = 5.0, 3.8 Hz, 1H), 6.86 (dd, J = 10.8, 6.4 Hz, 1H), 6.75 (d, J = 6.4 Hz, 1H), 6.66 (dd, J = 10.8, 5.8 Hz, 1H), 6.14 (dd, J = 8.6, 5.8 Hz, 1H), 5.32 (dd, J = 11.6, 4.4 Hz, 1H), 5.16 (dd, J = 11.6, 3.8 Hz, 1H), 4.29 (t, J = 8.6 Hz, 1H), 4.22−4.08 (m, 4H), 2.82 (dd, J = 14.0, 1.7 Hz, 1H), 2.49 (dt, J = 14.0, 4.2 Hz, 1H), 1.76 (dd, J = 7.4, 1.5 Hz, 1H), 1.23 (t, J = 7.1 Hz, 3H), 1.13 (t, J = 7.1 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 207.7, 169.3, 168.8, 136.7, 134.1, 132.7, 129.4, 127.5, 127.0, 126.5, 116.7, 114.2, 114.1, 102.1, 81.5, 61.9, 61.8, 55.9, 40.3, 30.5, 14.1, 14.0; IR (neat): ν 2983, 1960, 1733, 1443, 1368, 1259, 1164, 1015, 715 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{22}H_{23}NO_6NaS_2$ 484.0859; found 484.0851.

*1-(Naphthalen-2-ylsulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diethyl (**3q**).* Yield: 0.0339 g, 67%; a colorless solid; mp: 150−154 °C; Eluent:PE/EA = 4:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 7.90−7.81 (m, 2H), 7.78 (d, J = 8.0 Hz, 1H), 7.72 (s, 1H), 7.64 (t, J = 7.6 Hz, 1H), 7.58 (t, J = 7.6 Hz, 1H), 7.56−7.49 (m, 1H), 6.93−6.73 (m, 2H), 6.56 (dd, J = 10.2, 5.8 Hz, 1H), 5.63 (dd, J = 8.2, 2.0 Hz, 1H), 5.34 (dd, J = 11.6, 4.4 Hz, 1H), 5.18 (dd, J = 11.6, 3.8 Hz, 1H), 4.24−4.11 (m, 2H), 4.08−3.98 (m, 2H), 3.79 (t, J = 8.2 Hz, 1H), 2.78 (dd, J = 13.9, 1.6 Hz, 1H), 2.34 (dt, J = 13.9, 4.2 Hz, 1H), 1.66 (d, J = 7.6 Hz, 1H), 1.22 (t, J = 7.1 Hz, 3H), 1.04 (t, J = 7.1 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 207.7, 169.3, 168.7, 134.8, 133.3, 131.5, 130.3, 129.4, 129.2, 129.0, 128.9, 127.8, 127.6, 127.2, 126.1, 122.5, 116.9, 114.0, 113.8, 102.1, 81.3, 61.9, 61.7, 55.8, 40.4, 30.6, 14.1, 13.9; IR (neat): ν 3040, 2943, 2012, 1735, 1503, 1438, 1364, 1259, 1165, 1132, 1091, 713 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + H]^+$ Calcd for $C_{28}H_{28}NO_6S$ 506.1632; found 506.1635.

*1-(Methylsulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diethyl (**3s**).* Yield: 0.0368 g, 94%; a colorless solid; mp: 110−114 °C; Eluent:PE/EA = 4:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 6.82 (dd, J = 11.0, 6.4 Hz, 1H), 6.70 (dd, J = 11.0, 5.8 Hz, 1H), 6.49 (d, J = 6.4 Hz, 1H), 6.44 (t, J = 7.4 Hz, 1H), 5.28 (dd, J = 11.2, 3.8 Hz, 1H), 5.17 (dd, J = 11.2, 4.2 Hz 1H), 5.10 (t, J = 8.1 Hz, 1H), 4.23 (q, J = 7.5 Hz, 4H), 3.09−2.90 (m, 2H), 2.76 (s, 3H), 2.01 (d, J = 7.4 Hz, 1H), 1.29−1.19 (m, 6H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 205.9, 169.4, 168.9, 129.5, 127.6, 127.2, 118.8, 116.6, 113.1, 102.4, 82.2, 62.0, 61.9, 56.1, 41.2, 38.4, 31.8, 14.1, 14.0; IR (neat): ν −2985, 2930, 2041, 1733, 1443, 1356, 1260, 1160, 1033, 960, 715 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{19}H_{23}NO_6NaS$ 416.1138; found 416.1128.

*1-(Ethylsulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diethyl (**3t**).* Yield: 0.0348 g, 92%; a colorless solid; mp: 111−114 °C; Eluent:PE/EA = 4:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 6.80 (dd, J = 10.8, 6.4 Hz, 1H), 6.68 (dd, J = 10.8, 5.6 Hz, 1H), 6.47 (d, J = 6.4 Hz, 1H), 6.43 (dd, J = 8.9, 5.6 Hz, 1H), 5.26 (dd, J = 11.2, 3.4 Hz, 1H), 5.15 (dd, J = 11.2, 3.0 Hz, 1H), 5.07 (t, J = 8.1 Hz, 1H), 4.30−4.13 (m, 4H), 3.08−2.97 (m, 2H), 2.90 (qd, J = 7.2, 3.6 Hz, 2H), 2.02 (dt, J = 7.5, 1.3 Hz, 1H), 1.33−1.17 (m, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 205.7, 169.4, 168.9, 129.5, 127.2, 127.1, 118.9, 116.5, 112.0, 102.4, 82.1, 62.0, 61.9, 56.3, 44.9, 41.1, 32.0, 14.1, 14.0, 7.7; IR (neat): ν 2980, 2022, 1734, 1440, 1350, 1260, 1185, 1134, 1028, 718 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{20}H_{25}NO_6NaS$ 430.1295; found 430.1294.

*1-(Isopropylsulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diethyl (**3u**).* Yield: 0.0296 g, 68%; a colorless solid; mp: 112−115 °C; Eluent:PE/EA = 4:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 6.79 (dd, J = 10.8, 6.4 Hz, 1H), 6.66 (dd, J = 10.8, 5.8 Hz, 1H), 6.47−6.38 (m, 2H), 5.24 (dd, J = 11.2, 3.6 Hz, 1H), 5.14 (dd, J = 11.2, 3.2 Hz, 1H), 5.05 (t, J = 8.2 Hz, 1H), 4.29−4.12 (m, 4H), 3.21 (hept, J = 6.8 Hz, 1H), 3.11−2.97 (m, 2H), 2.04 (dt, J = 7.6, 1.1 Hz, 1H), 1.34−1.14 (m, 12H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 205.5, 169.5, 168.9, 129.5, 127.0, 126.9, 119.2, 116.4, 111.2, 102.7, 82.0, 62.0, 61.9, 56.4, 50.5, 41.0, 31.9, 16.7, 15.9, 14.1, 14.0; IR (neat): ν 2980, 2936, 1960, 1734, 1443, 1350, 1260, 1183, 1147, 1134, 1031, 715 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{21}H_{27}NO_6NaS$ 444.1451; found 444.1443.

*1-(Cyclopropylsulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diethyl (**3v**).* Yield: 0.0404 g, 96%; a colorless solid; mp: 114−116 °C; Eluent:PE/EA = 4:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 6.82 (dd, J = 10.8, 6.4 Hz, 1H), 6.69 (dd, J = 10.8, 5.8 Hz, 1H), 6.59 (d, J = 6.4 Hz, 1H), 6.41 (dd, J = 8.9, 5.8 Hz, 1H), 5.22 (d, J = 10.8 Hz, 1H), 5.13 (d, J = 10.8 Hz, 1H), 5.05 (t, J = 8.2 Hz, 1H), 4.27−4.17 (m, 4H), 3.02 (s, 2H), 2.26−2.16 (m, 1H), 2.00 (d, J = 7.5 Hz, 1H), 1.29−1.17 (m, 6H), 1.12−0.99 (m, 2H), 0.89−0.82 (m, 2H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 206.6, 169.4, 169.0, 129.6, 127.2, 126.8, 118.6, 116.3, 113.0, 102.3, 81.5, 62.0, 61.9, 56.2, 40.9, 31.7, 28.8, 14.1, 14.0, 6.4, 5.4; IR (neat): ν 2985, 2923, 1994, 1733, 1443, 1357, 1259, 1155, 1031, 885, 718 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{21}H_{25}NO_6NaS$ 442.1295; found 442.1291.

*1-(tert-Butylsulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diethyl (**3w**).* Yield: 0.0137 g, 32%; a pale-yellow solid; mp: 113−120 °C; Eluent:PE/EA = 4:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 6.76 (dd, J = 10.6, 6.4 Hz, 1H), 6.71 (d, J = 6.4 Hz, 1H), 6.64 (dd, J = 10.6, 5.8 Hz, 1H), 6.43 (dd, J = 8.8, 5.8 Hz, 1H), 5.17 (dd, J = 11.2, 3.7 Hz, 1H), 5.13−5.04 (m, 2H), 4.30−4.14 (m, 4H), 3.13 (dt, J = 13.8, 3.5 Hz, 1H), 3.02 (dd, J = 13.8, 1.5 Hz, 1H), 2.03 (d, J = 7.8 Hz, 1H), 1.34−1.16 (m,

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15H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 206.6, 169.6, 169.0, 129.7, 126.9, 126.8, 120.5, 117.2, 112.1, 104.0, 81.4, 65.1, 61.9, 61.8, 56.9, 41.5, 32.7, 25.5, 14.1, 14.0; IR (neat): ν 2967, 2909, 1957, 1735, 1438, 1365, 1259, 1168, 1031, 715 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{22}H_{29}NO_6NaS$ 458.1608; found 458.1614.

*1-(Methylsulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Dimethyl (**3y**).* Yield: 0.0337 g, 92%; a colorless solid; mp: 109−110 °C; Eluent:PE/EA = 4:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 6.82 (dd, J = 10.9, 6.2 Hz, 1H), 6.70 (dd, J = 10.9, 5.6 Hz, 1H), 6.48 (d, J = 6.2 Hz, 1H), 6.44 (dd, J = 8.9, 5.6 Hz, 1H), 5.28 (dd, J = 11.4, 3.6 Hz, 1H), 5.19 (dd, J = 11.4, 3.2 Hz, 1H), 5.09 (t, J = 8.1 Hz, 1H), 3.75 (s, 6H), 3.08−2.90 (m, 2H), 2.76 (s, 3H), 2.02 (d, J = 7.3 Hz, 1H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 205.7, 169.8, 169.3, 129.5, 127.6, 127.3, 118.7, 116.4, 113.1, 102.2, 82.2, 56.3, 53.0, 41.2, 38.4, 31.9; IR (neat): ν 3019, 2951, 1970, 1736, 1438, 1355, 1266, 1160, 960, 715 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{17}H_{19}NO_6NaS$ 388.0825; found 388.0831.

*1-(Methylsulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Diisopropyl (**3z**).* Yield: 0.0372 g, 88%; a colorless solid; mp: 112−113 °C; Eluent:PE/EA = 4:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 6.82 (dd, J = 10.8, 6.4 Hz, 1H), 6.70 (dd, J = 10.8, 5.6 Hz, 1H), 6.50 (d, J = 6.4 Hz, 1H), 6.43 (dd, J = 8.9, 5.6 Hz, 1H), 5.27 (dd, J = 11.2, 3.7 Hz, 1H), 5.19−5.00 (m, 4H), 3.04−2.90 (m, 2H), 2.75 (s, 3H), 1.98 (dt, J = 7.4, 1.3 Hz, 1H), 1.28−1.14 (m, 12H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 206.0, 168.9, 168.4, 129.5, 127.5, 127.0, 118.9, 116.7, 113.0, 102.5, 82.3, 69.7, 69.6, 55.9, 41.1, 38.4, 31.7, 21.6, 21.53, 21.47, 21.4; IR (neat): ν 2980, 2928, 2020, 1729, 1435, 1357, 1266, 1161, 1106, 966, 713 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{21}H_{27}NO_6NaS$ 444.1451; found 444.1453.

*1-(Methylsulfonyl)-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Dibenzyl (**3aa**).* Yield: 0.0437 g, 84%; a colorless solid; mp: 127−132 °C; Eluent:PE/EA = 4:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 7.33−7.21 (m, 10H), 6.80 (dd, J = 10.9, 6.2 Hz, 1H), 6.67 (dd, J = 10.9, 5.8 Hz, 1H), 6.47 (d, J = 6.4 Hz, 1H), 6.33 (dd, J = 8.9, 5.8 Hz, 1H), 5.18−5.00 (m, 6H), 4.81 (dd, J = 11.2, 3.2 Hz, 1H), 3.08−2.92 (m, 2H), 2.72 (s, 3H), 2.03 (d, J = 7.4 Hz, 1H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 205.6, 169.1, 168.6, 135.0, 134.9, 129.5, 128.6, 128.5, 128.45, 128.4, 128.3, 127.6, 127.3, 127.0, 118.6, 116.4, 113.1, 101.9, 82.2, 67.6, 67.5, 56.3, 41.3, 38.4, 31.8; IR (neat): ν 2967, 1968, 1735, 1438, 1357, 1261, 1162, 958, 699 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{29}H_{27}NO_6NaS$ 540.1451; found 540.1450.

*2-(1λ5-But-1-en-1-ylidene)-1-(methylsulfonyl)-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate Dimethyl (**3ab**).* Yield: 0.0284 g, 72%; a colorless solid; mp: 110−113 °C; Eluent:PE/EA = 4:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 6.78 (dd, J = 10.9, 6.4 Hz, 1H), 6.67 (dd, J = 10.9, 5.6 Hz, 1H), 6.48−6.37 (m, 2H), 5.59 (td, J = 6.8, 2.6 Hz, 1H), 5.07 (t, J = 8.1 Hz, 1H), 3.72 (s, 3H), 3.71 (s, 3H), 3.03−2.90 (m, 2H), 2.73 (s, 3H), 2.15 (qd, J = 7.5, 4.4 Hz, 2H), 1.97 (d, J = 7.4 Hz, 1H), 1.07 (t, J = 7.5 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 200.0, 170.0, 169.4, 129.5, 127.4, 127.2, 119.2, 116.5, 113.0, 101.7, 100.3, 56.4, 53.0, 52.9, 41.5, 38.2, 32.7, 22.4, 13.2; IR (neat): ν 2954, 1986, 1738, 1435, 1354, 1265, 1158, 963, 749 cm^{-1}; HRMS (ESI-FTMS) m/z: $[M + H]^+$ Calcd for $C_{19}H_{24}NO_6S$ 394.1319; found 394.1311.

*Diethyl 10-(Dicyanomethylene)-9-vinylidenebicyclo[4.3.1]deca-2,4-diene-7,7-dicarboxylate (**3ae**).* Yield: 0.0299 g, 82%; a colorless solid; mp: 184−186 °C; Eluent:PE/EA = 4:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 6.11 (dd, J = 11.2, 7.8 Hz, 1H), 6.01 (dd, J = 11.2, 7.8 Hz, 1H), 5.84 (dd, J = 11.2, 7.8 Hz, 1H), 5.59 (dd, J = 11.2, 7.8 Hz, 1H), 4.86 (dd, J = 11.2, 4.4 Hz, 1H), 4.79 (dd, J = 11.2, 4.2 Hz, 1H), 4.70 (dd, J = 8.0, 2.0 Hz, 1H), 4.48 (dd, J = 8.0, 2.6 Hz, 1H), 4.36−4.12 (m, 4H), 2.78 (dt, J = 14.4, 1.0 Hz, 1H), 2.67 (dt, J = 14.0, 4.2 Hz, 1H), 1.30 (t, J = 7.2 Hz, 3H), 1.25 (t, J = 7.2 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 205.2, 174.6, 168.5, 167.3, 129.1, 127.4, 124.8, 122.6, 111.2, 111.1, 95.9, 85.4, 77.2, 62.4, 62.37, 62.34, 48.3, 47.0, 28.2, 14.1, 14.0; IR (neat): ν 2956, 2920, 2870, 2850, 2234, 1960, 1736, 1594, 1495, 1465, 1362, 1300, 1255, 1234, 1204, 1081, 1042, 859, 697 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{21}H_{20}N_2O_4Na$ 387.1315; found 387.1315.

*General Procedure for the Synthesis of Compounds **5**.* Tris(dibenzylideneacetone)dipalladium (0.0023 g, 0.002 mmol, 2.5 mol %) and DPEPhos (0.0040 g, 0.007 mmol, 7.5 mol %) were added to an oven-dried 10 mL sealed tube equipped with a magnetic stirring bar under an argon atmosphere. The resulting catalyst system was stirred in THF (degassed) (1.0 mL) for 30 min. After that, **4** (0.1 mmol), **2** (0.2 mmol), $Yb(OTf)_3$ (0.0062 g, 0.01 mmol, 10 mol %), and Cs_2CO_3 (0.0325 g, 0.1 mmol, 1.0 equiv) were added under an argon atmosphere. The reaction mixture was stirred at 50 °C in an oil bath for 24 h. The reaction solution was filtered through a pad of Celite to remove insoluble solids. Then, the solvent was evaporated under vacuum, and the residue was purified by column chromatography on silica gel (petroleum ether/ethyl acetate = 10/1) to afford the desired product **5** as a white solid.

*9-Phenyl-2-vinylcyclohepta[b]pyran-4,4(4aH)-dicarboxylate Diethyl (**5a**).* Yield: 0.0264 g, 67%; a colorless solid; mp: 104−106 °C; Eluent:PE/EA = 10:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 7.47 (d, J = 7.6 Hz, 2H), 7.35 (t, J = 7.6 Hz, 2H), 7.25 (t, J = 7.2 Hz, 1H), 6.89 (d, J = 11.0 Hz, 1H), 6.81 (dd, J = 11.0, 5.2 Hz, 1H), 6.35 (dd, J = 8.8, 5.4 Hz, 1H), 6.13 (dd, J = 17.2, 11.0 Hz, 1H), 5.65 (s, 1H), 5.31 (d, J = 17.2 Hz, 1H), 5.18 (dd, J = 8.8, 6.8 Hz, 1H), 5.08 (d, J = 11.0 Hz, 1H), 4.30 (q, J = 7.1 Hz, 2H), 4.26−4.07 (m, 2H), 2.33 (d, J = 6.8 Hz, 1H), 1.28−1.22 (m, 6H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 168.7, 168.5, 150.4, 137.3, 134.8, 131.5, 130.4, 129.4, 128.2, 127.7, 126.6, 126.3, 116.9, 116.8, 116.7, 98.7, 62.1, 61.9, 53.6, 39.7, 14.1, 14.0; IR (neat): ν 2979, 2926, 1733, 1659, 1597, 1488, 1364, 1334, 1250, 1206, 1092, 830 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{24}H_{24}O_5Na$ 415.1516; found 415.1514.

*9-(p-Tolyl)-2-vinylcyclohepta[b]pyran-4,4(4aH)-dicarboxylate Diethyl (**5b**).* Yield: 0.0215 g, 53%; a colorless solid; mp: 103−107 °C; Eluent:PE/EA = 10:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 7.36 (d, J = 8.2 Hz, 2H), 7.16 (d, J = 8.2 Hz, 2H), 6.87 (d, J = 11.0 Hz, 1H), 6.79 (dd, J = 11.0, 5.2 Hz, 1H), 6.34 (ddd, J = 8.8, 5.2, 1.2 Hz, 1H), 6.13 (dd, J = 17.0, 11.0 Hz, 1H), 5.64 (d, J = 1.4 Hz, 1H), 5.34 (dd, J = 17.0, 1.4 Hz, 1H), 5.17 (dd, J = 8.8, 7.0 Hz, 1H), 5.09 (dd, J = 11.0, 1.0 Hz, 1H), 4.32−4.09 (m, 4H), 2.36 (s, 3H), 2.31 (dt, J = 7.0, 1.4 Hz, 1H), 1.28−1.22 (m, 6H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 168.8, 168.5, 150.5, 136.2, 134.6, 134.4, 131.7, 130.3, 129.5, 128.5, 128.1, 126.2, 116.8, 116.7, 98.7, 62.0, 61.9, 53.6, 39.8, 21.2, 14.1, 14.0; IR (neat): ν 2979, 2957, 2925, 1734, 1664, 1601, 1513, 1460, 1253, 1206, 1027, 819 cm^{-1};

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HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{25}H_{26}O_5Na$ 429.1673; found 429.1668.

*9-(4-Chlorophenyl)-2-vinylcyclohepta[b]pyran-4,4(4aH)-dicarboxylate Diethyl (**5c**).* Yield: 0.0294 g, 69%; a colorless solid; mp: 109−112 °C; Eluent:PE/EA = 10:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 7.41 (d, J = 8.6 Hz, 2H), 7.32 (d, J = 8.6 Hz, 2H), 6.82 (d, J = 2.8 Hz, 2H), 6.35 (dt, J = 9.0, 2.8 Hz, 1H), 6.13 (dd, J = 17.2, 11.0 Hz, 1H), 5.66 (s, 1H), 5.30 (d, J = 17.2 Hz, 1H), 5.18 (dd, J = 8.8, 6.8 Hz, 1H), 5.11 (d, J = 11.0 Hz, 1H), 4.31−4.11 (m, 4H), 2.32 (d, J = 6.8 Hz, 1H), 1.28−1.22 (m, 6H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 168.6, 168.4, 150.4, 135.8, 135.2, 132.4, 131.7, 131.0, 129.4, 128.5, 127.9, 126.4, 117.0, 116.7, 115.8, 99.0, 62.1, 62.0, 53.5, 39.8, 14.1, 14.0; IR (neat): ν 2985, 2935, 1733, 1667, 1600, 1516, 1331, 1251, 1207, 1026, 734 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{24}H_{23}O_5NaCl$ 449.1126; found 449.1124.

*9-(4-Methoxyphenyl)-2-vinylcyclohepta[b]pyran-4,4-(4aH)-dicarboxylate Diethyl (**5d**).* Yield: 0.0173 g, 41%; a colorless solid; mp: 87−90 °C; Eluent:PE/EA = 10:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 7.40 (d, J = 8.8 Hz, 2H), 6.90 (d, J = 8.8 Hz, 2H), 6.86 (d, J = 11.0 Hz, 1H), 6.79 (dd, J = 11.0, 5.2 Hz, 1H), 6.34 (ddd, J = 8.8, 5.2, 1.2 Hz, 1H), 6.13 (dd, J = 17.0, 11.0 Hz, 1H), 5.64 (d, J = 1.4 Hz, 1H), 5.34 (dd, J = 17.0, 1.4 Hz, 1H), 5.17 (dd, J = 8.8, 6.8 Hz, 1H), 5.09 (dd, J = 10.8, 1.6 Hz, 1H), 4.34−4.11 (m, 4H), 3.83 (s, 3H), 2.31 (d, J = 6.8 Hz, 1H), 1.27−1.23 (m, 6H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 168.8, 168.6, 158.3, 150.5, 134.4, 131.7, 131.6, 129.9, 129.5, 128.1, 126.2, 116.8, 116.6, 116.5, 113.2, 98.7, 62.0, 61.9, 55.2, 53.6, 39.8, 14.1, 14.0; IR (neat): ν 2979, 2957, 2923, 2853, 1732, 1659, 1603, 1511, 1457, 1336, 1244, 1205, 1029, 832 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{25}H_{26}O_6Na$ 445.1621; found 445.1619.

*9-(4-Formylphenyl)-2-vinylcyclohepta[b]pyran-4,4(4aH)-dicarboxylate Diethyl (**5e**).* Yield: 0.0336 g, 80%; a colorless solid; mp: 105−108 °C; Eluent:PE/EA = 10:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 10.02 (s, 1H), 7.87 (d, J = 8.2 Hz, 2H), 7.66 (d, J = 8.2 Hz, 2H), 6.92−6.81 (m, 2H), 6.38 (dddd, J = 8.8, 3.4, 2.0, 1.4 Hz, 1H), 6.14 (dd, J = 17.2, 11.0 Hz, 1H), 5.69 (d, J = 1.4 Hz, 1H), 5.28 (d, J = 17.2 Hz, 1H), 5.21 (dd, J = 8.8, 6.8 Hz, 1H), 5.11 (dt, J = 11.0, 0.8 Hz, 1H), 4.32−4.11 (m, 4H), 2.38 (dt, J = 6.8, 1.4 Hz, 1H), 1.29−1.23 (m, 6H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 192.1, 168.4, 168.3, 150.2, 143.9, 136.4, 134.5, 131.0, 130.5, 129.3, 129.2, 128.9, 126.6, 117.1, 116.8, 115.8, 99.2, 62.2, 62.0, 53.5, 39.9, 14.1, 14.0; IR (neat): ν 2979, 2932, 2853, 2733, 1732, 1700, 1601, 1561, 1446, 1253, 1206, 1168, 1025, 832 cm^{-1}; HRMS (ESI-FTMS) m/z: $[M + H]^+$ Calcd for $C_{25}H_{25}O_6$ 421.1646; found 421.1648.

*9-(4-(Methoxycarbonyl)phenyl)-2-vinylcyclohepta[b]-pyran-4,4(4aH)-dicarboxylate Diethyl (**5f**).* Yield: 0.0324 g, 72%; a colorless solid; mp: 106−108 °C; Eluent:PE/EA = 10:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 8.02 (d, J = 8.4 Hz, 2H), 7.55 (d, J = 8.4 Hz, 2H), 6.90−6.82 (m, 2H), 6.37 (ddt, J = 8.8, 4.4, 1.2 Hz, 1H), 6.13 (dd, J = 17.2, 11.0 Hz, 1H), 5.68 (d, J = 1.4 Hz, 1H), 5.29 (dd, J = 17.2, 1.4 Hz, 1H), 5.20 (dd, J = 8.8, 6.8 Hz, 1H), 5.10 (dd, J = 10.8, 1.4 Hz, 1H), 4.33−4.10 (m, 4H), 3.93 (s, 3H), 2.36 (d, J = 6.8 Hz, 1H), 1.29−1.22 (m, 6H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 168.5, 168.4, 167.2, 150.3, 142.2, 136.0, 130.8, 130.4, 129.3, 129.1, 128.7, 128.1, 126.5, 117.0, 116.8, 116.0, 99.0, 62.2, 62.0, 53.5, 52.0, 39.8, 14.1, 14.0; IR (neat): ν 3086, 2979, 2954, 1723, 1659, 1600, 1502, 1435, 1277, 1253, 1204, 1102, 1024, 775 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{26}H_{26}O_7Na$ 473.1571; found 473.1566.

*9-(4-Nitrophenyl)-2-vinylcyclohepta[b]pyran-4,4(4aH)-dicarboxylate Diethyl (**5g**).* Yield: 0.0315 g, 72%; a colorless solid; mp: 115−118 °C; Eluent:PE/EA = 10:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 8.21 (d, J = 8.8 Hz, 2H), 7.65 (d, J = 8.8 Hz, 2H), 6.91−6.82 (m, 2H), 6.39 (dd, J = 8.8, 5.0 Hz, 1H), 6.15 (dd, J = 17.2, 11.0 Hz, 1H), 5.76−5.68 (m, 1H), 5.30−5.19 (m, 2H), 5.13 (d, J = 11.0 Hz, 1H), 4.33−4.12 (m, 4H), 2.39 (d, J = 6.8 Hz, 1H), 1.28−1.24 (m, 6H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 168.3, 168.2, 150.2, 146.2, 144.3, 136.9, 131.0, 130.0, 129.3, 129.1, 126.8, 123.1, 117.2, 116.9, 114.9, 99.4, 62.3, 62.1, 53.4, 39.9, 14.1, 14.0; IR (neat): ν 2979, 2923, 2853, 1732, 1662, 1597, 1516, 1451, 1345, 1300, 1206, 1026, 858 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{24}H_{23}NO_7Na$ 460.1367; found 460.1359.

*9-(4-(Trifluoromethyl)phenyl)-2-vinylcyclohepta[b]pyran-4,4(4aH)-dicarboxylate Diethyl (**5h**).* Yield: 0.0327 g, 71%; a colorless solid; mp: 105−107 °C; Eluent:PE/EA = 10:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 7.60 (d, J = 9.2 Hz, 2H), 7.58 (d, J = 9.2 Hz, 2H), 6.87−6.82 (m, 2H), 6.37 (dddd, J = 8.8, 3.6, 2.4, 1.2 Hz, 1H), 6.14 (dd, J = 17.2, 11.0 Hz, 1H), 5.70−5.67 (m, 1H), 5.28 (ddd, J = 17.2, 1.2, 0.8 Hz, 1H), 5.20 (dd, J = 8.8, 6.8 Hz, 1H), 5.11 (ddd, J = 11.0, 1.2, 0.8 Hz, 1H), 4.32−4.11 (m, 4H), 2.36 (dt, J = 6.8, 1.2 Hz, 1H), 1.29−1.22 (m, 6H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 168.5, 168.4, 150.3, 141.0, 135.9, 130.7, 130.6, 129.3, 128.7, 128.5 (q, J = 32.3 Hz), 126.5, 124.6 (q, J = 3.8 Hz), 124.4 (q, J = 272.8 Hz), 117.0, 116.8, 115.6, 99.1, 62.2, 62.0, 53.5, 39.8, 14.1, 14.0; ^{19}F NMR (376 MHz, $CDCl_3$) δ −62.32; IR (neat): ν 2985, 2937, 2907, 1733, 1667, 1614, 1597, 1407, 1324, 1254, 1206, 1162, 1123, 1067, 843 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{25}H_{23}O_5F_3Na$ 483.1390; found 483.1391.

*9-(3-Cyanophenyl)-2-vinylcyclohepta[b]pyran-4,4(4aH)-dicarboxylate Diethyl (**5i**).* Yield: 0.0271 g, 65%; a colorless solid; mp: 107−110 °C; Eluent:PE/EA = 10:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 7.77 (td, J = 1.8, 0.6 Hz, 1H), 7.73 (dt, J = 7.8, 1.6 Hz, 1H), 7.54 (dt, J = 7.8, 1.4 Hz, 1H), 7.46 (td, J = 7.8, 0.6 Hz, 1H), 6.86 (dd, J = 11.0, 5.0 Hz, 1H), 6.80 (d, J = 11.0 Hz, 1H), 6.38 (ddd, J = 8.8, 5.0, 1.4 Hz, 1H), 6.14 (dd, J = 17.2, 11.0 Hz, 1H), 5.70 (d, J = 1.4 Hz, 1H), 5.28−5.17 (m, 2H), 5.12 (dt, J = 11.0, 1.0 Hz, 1H), 4.33−4.12 (m, 4H), 2.36 (dt, J = 7.0, 1.4 Hz, 1H), 1.29−1.23 (m, 6H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 168.4, 168.3, 150.2, 138.6, 136.0, 134.8, 134.0, 130.2, 130.1, 129.2, 129.1, 128.6, 126.6, 119.0, 117.1, 116.7, 114.8, 112.0, 99.3, 62.2, 62.0, 53.4, 39.8, 14.1, 14.0; IR (neat): ν 2979, 2926, 2228, 1731, 1664, 1600, 1443, 1370, 1334, 1245, 1206, 1161, 1026, 797 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{25}H_{23}NO_5Na$ 440.1468; found 440.1464.

*9-(Naphthalen-2-yl)-2-vinylcyclohepta[b]pyran-4,4(4aH)-dicarboxylate Diethyl (**5k**).* Yield: 0.0279 g, 63%; a colorless solid; mp: 104−109 °C; Eluent:PE/EA = 10:1. ^{1}H NMR (400 MHz, $CDCl_3$, TMS) δ 7.86−7.74 (m, 5H), 7.49−7.42 (m, 2H), 6.99 (d, J = 11.0 Hz, 1H), 6.87 (dd, J = 11.0, 5.2 Hz, 1H), 6.40 (ddd, J = 8.8, 5.2, 1.2 Hz, 1H), 6.13 (dd, J = 17.2, 11.0 Hz, 1H), 5.68 (d, J = 1.4 Hz, 1H), 5.28 (dd, J = 17.2, 1.2 Hz, 1H), 5.22 (dd, J = 8.8, 6.8 Hz, 1H), 5.04 (dt, J = 11.0, 0.8 Hz, 1H), 4.34−4.12 (m, 4H), 2.39 (dt, J = 6.8, 1.4 Hz, 1H), 1.30−1.24 (m, 6H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 168.7, 168.6, 150.4, 135.2, 135.1, 133.4, 132.3, 131.6, 129.4, 129.2, 129.0, 128.4, 128.0, 127.5, 126.9, 126.4, 125.8, 125.6, 116.9, 116.8, 116.7, 98.8, 62.1, 62.0, 53.6, 39.8, 14.1, 14.0; IR

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(neat): ν 2957, 2923, 2853, 1733, 1667, 1600, 1463, 1446, 1334, 1256, 1206, 1026, 748 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{28}H_{26}O_5Na$ 465.1673; found 465.1678.

*9-(Thiophen-2-yl)-2-vinylcyclohepta[b]pyran-4,4(4aH)-dicarboxylate Diethyl (**5l**).* Yield: 0.0191 g, 48%; a colorless solid; mp: 94−98 °C; Eluent:PE/EA = 10:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 7.29 (dd, J = 5.2, 1.2 Hz, 1H), 7.22 (dd, J = 3.6, 1.2 Hz, 1H), 7.11 (dd, J = 11.0, 0.8 Hz, 1H), 7.04 (dd, J = 5.2, 3.6 Hz, 1H), 6.80 (dd, J = 11.0, 5.2 Hz, 1H), 6.33 (ddd, J = 8.8, 5.2, 1.2 Hz, 1H), 6.21 (dd, J = 17.0, 11.0 Hz, 1H), 5.81 (dd, J = 17.0, 1.2 Hz, 1H), 5.70 (d, J = 1.4 Hz, 1H), 5.30−5.22 (m, 2H), 4.32−4.07 (m, 4H), 2.41 (dt, J = 6.8, 1.4 Hz, 1H), 1.27−1.19 (m, 6H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 168.5, 168.3, 150.9, 139.0, 136.3, 130.2, 129.4, 128.7, 126.6, 126.5, 125.9, 124.8, 118.6, 117.6, 110.3, 99.4, 62.1, 62.0, 53.4, 39.8, 14.1, 14.0; IR (neat): ν 2982, 2926, 2850, 1733, 1664, 1606, 1465, 1446, 1334, 1281, 1256, 1203, 1025, 700 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{22}H_{22}O_5NaS$ 421.1080; found 421.1080.

*9-(4-Formylphenyl)-2-vinylcyclohepta[b]pyran-4,4(4aH)-dicarboxylate Dimethyl (**5o**).* Yield: 0.0282 g, 63%; a colorless solid; mp: 106−109 °C; Eluent:PE/EA = 10:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 10.02 (s, 1H), 7.87 (d, J = 8.2 Hz, 2H), 7.66 (d, J = 8.2 Hz, 2H), 6.91−6.81 (m, 2H), 6.37 (ddt, J = 8.8, 3.8, 1.4 Hz, 1H), 6.14 (dd, J = 17.2, 11.0 Hz, 1H), 5.68 (d, J = 1.4 Hz, 1H), 5.28 (dd, J = 17.2, 1.4 Hz, 1H), 5.23−5.00 (m, 4H), 2.35 (dt, J = 6.8, 1.4 Hz, 1H), 1.26−1.21 (m, 12H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 192.1, 167.9, 167.8, 150.2, 143.9, 136.6, 134.5, 131.0, 130.4, 129.3, 129.2, 128.8, 126.5, 117.2, 116.6, 115.7, 99.4, 69.8, 69.7, 53.5, 39.9, 21.6, 21.5, 21.4; IR (neat): ν 2982, 2929, 2736, 1727, 1701, 1664, 1602, 1561, 1465, 1451, 1258, 1209, 1104, 919, 830 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{27}H_{28}O_6Na$ 471.1778; found 471.1776.

*9-(4-Formylphenyl)-2-vinylcyclohepta[b]pyran-4,4(4aH)-dicarboxylate Dibenzyl (**5p**).* Yield: 0. 0392 g, 72%; a colorless solid; mp: 117−119 °C; Eluent:PE/EA = 10:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 10.01 (s, 1H), 7.82 (d, J = 8.2 Hz, 2H), 7.52 (d, J = 8.2 Hz, 2H), 7.34−7.27 (m, 7H), 7.25−7.18 (m, 3H), 6.87−6.79 (m, 2H), 6.29 (dddd, J = 8.8, 3.2, 2.4, 1.2 Hz, 1H), 6.13 (dd, J = 17.2, 11.0 Hz, 1H), 5.72 (d, J = 1.4 Hz, 1H), 5.28−5.03 (m, 7H), 2.41 (dt, J = 6.8, 1.4 Hz, 1H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 192.1, 168.2, 168.0, 150.5, 143.7, 136.1, 135.0, 134.9, 134.5, 131.0, 130.5, 129.2, 129.1, 128.9, 128.6, 128.5, 128.4 (2C), 128.3, 127.9, 126.7, 117.0, 116.9, 115.9, 98.8, 67.7, 53.8, 39.9; IR (neat): ν 3033, 2960, 2923, 2733, 1732, 1698, 1600, 1558, 1496, 1451, 1199, 1159, 833, 745 cm^{-1}; HRMS (ESI-FTMS) m/z: $[M + Na]^+$ Calcd for $C_{35}H_{29}O_6$ 545.1959; found 545.1961.

*9-(Tosyloxy)-2-vinylcyclohepta[b]pyran-4,4(4aH)-dicarboxylate Diethyl (**5q**).* Yield: 0.0360 g, 74%; a colorless solid; mp: 107−110 °C; Eluent:PE/EA = 10:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 7.72 (d, J = 8.2 Hz, 2H), 7.21 (d, J = 8.2 Hz, 2H), 6.71−6.61 (m, 2H), 6.29 (dddd, J = 8.8, 3.2, 2.0, 1.2 Hz, 1H), 6.01 (dd, J = 17.2, 11.0 Hz, 1H), 5.59−5.48 (m, 2H), 5.16 (d, J = 11.2 Hz, 1H), 5.10 (dd, J = 8.8, 6.8 Hz, 1H), 4.30−4.06 (m, 4H), 2.38 (s, 3H), 2.20 (d, J = 7.0 Hz, 1H), 1.27−1.16 (m, 6H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 168.1, 167.7, 149.9, 144.7, 133.3, 131.4, 129.4, 128.5, 128.4, 128.3, 128.0, 126.4, 125.2, 118.4, 117.3, 98.6, 62.2, 62.1, 52.7, 37.0, 21.6, 14.0, 13.9; IR (neat): ν 2979, 2923, 1732, 1595, 1460, 1364, 1248, 1206, 1177, 1087, 1032, 802, 777, 727 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{25}H_{26}O_8NaS$ 509.1241; found 509.1246.

*9-Chloro-2-vinylcyclohepta[b]pyran-4,4(4aH)-dicarboxylate Diethyl (**5r**).* Yield: 0.0274 g, 78%; a colorless solid; mp: 87−89 °C; Eluent:PE/EA = 10:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 6.76 (d, J = 11.2 Hz, 1H), 6.65 (dd, J = 11.2, 5.4 Hz, 1H), 6.28 (dd, J = 9.0, 5.4 Hz, 1H), 6.19 (dd, J = 17.2, 11.0 Hz, 1H), 5.84 (d, J = 17.2 Hz, 1H), 5.67 (s, 1H), 5.30 (d, J = 11.0 Hz, 1H), 5.15 (t, J = 8.0 Hz, 1H), 4.30−4.07 (m, 4H), 2.43 (d, J = 6.8 Hz, 1H), 1.27−1.17 (m, 6H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 168.2, 167.9, 150.5, 134.7, 129.4, 129.1, 128.1, 126.3, 118.1, 117.3, 108.3, 99.2, 62.2, 62.1, 53.2, 39.1, 14.0, 13.9; IR (neat): ν 2980, 2938, 1732, 1662, 1600, 1367, 1333, 1250, 1206, 1160, 1025, 734 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{18}H_{19}O_5NaCl$ 373.0813; found 373.0808.

*9-Chloro-7-(prop-1-en-2-yl)-2-vinylcyclohepta[b]pyran-4,4(4aH)-dicarboxylate Diethyl (**5s**).* Yield: 0.0281 g, 72%; a colorless solid; mp: 105−108 °C; Eluent:PE/EA = 10:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 6.79 (s, 1H), 6.38 (d, J = 9.2 Hz, 1H), 6.20 (dd, J = 17.2, 11.0 Hz, 1H), 5.86 (dd, J = 17.2, 1.2 Hz, 1H), 5.65 (d, J = 1.2 Hz, 1H), 5.31 (d, J = 11.0 Hz, 1H), 5.24 (dd, J = 9.2, 7.4 Hz, 1H), 5.14 (s, 2H), 4.36−4.06 (m, 4H), 2.61 (dq, J = 7.4, 1.2 Hz, 1H), 2.08 (s, 3H), 1.27−1.15 (m, 6H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 168.1, 168.0, 150.5, 142.8, 141.0, 137.0, 129.1, 125.6, 125.3, 118.8, 117.3, 116.1, 108.7, 99.1, 62.3, 62.0, 53.0, 39.2, 20.7, 14.1, 13.9; IR (neat): ν 3092, 2982, 2929, 1735, 1667, 1601, 1463, 1446, 1331, 1253, 1208, 1028, 770 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{21}H_{23}O_5NaCl$ 413.1126; found 413.1130.

*9-Chloro-7-isopropyl-2-vinylcyclohepta[b]pyran-4,4-(4aH)-dicarboxylate Diethyl (**5t**).* Yield: 0.0287 g, 73%; a colorless solid; mp: 98−101 °C; Eluent:PE/EA = 10:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 6.54 (s, 1H), 6.25−6.12 (m, 2H), 5.84 (dd, J = 17.2, 1.2 Hz, 1H), 5.63 (d, J = 1.4 Hz, 1H), 5.28 (dd, J = 11.0, 1.4 Hz, 1H), 5.18 (dd, J = 9.2, 7.0 Hz, 1H), 4.37−4.07 (m, 4H), 2.64 (hept, J = 7.0 Hz, 1H), 2.49 (dq, J = 7.0, 1.2 Hz, 1H), 1.25−1.11 (m, 12H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 168.2, 168.1, 150.7, 149.0, 134.5, 129.2, 126.5, 123.9, 118.8, 117.2, 108.0, 98.9, 62.2, 61.9, 53.1, 39.3, 35.7, 23.3, 22.6, 14.0, 13.9; IR (neat): ν 2963, 2867, 1734, 1664, 1617, 1600, 1465, 1331, 1251, 1206, 1028, 764 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{21}H_{25}O_5NaCl$ 415.1283; found 415.1275.

*General Procedure for the Synthesis of **3a′**.* Compound **1-tosyl-2-(1λ5-vinylidene)-1,2,3,4a-tetrahydro-4*H*-cyclohepta-[*b*]pyridine-4,4-dicarboxylate diethyl 3a** (0.047 g, 0.10 mmol), Pd_2dba_3 (0.004 g, 0.004 mmol), and rac-BINAP (0.005 g, 0.008 mol) were added to an oven-dried 10 mL sealed tube equipped with a magnetic stirring bar under an argon atmosphere and added with THF (degassed) (0.5 mL) and DCE (degassed) (0.5 mL). Then, HCOOH (0.005 g, 0.11 mmol) was added. The reaction mixture was heated to 40 °C in an oil bath and stirred for 12 h. After that, the reaction mixture was then purified by flash chromatography on silica gel (petroleum ether/ethyl acetate = 4/1) to afford the desired product **3a′** (*trans* and *cis*-isomeric mixture) as a white solid in 71% yield (0.0335 g).

*Diethyl (E)-2-Ethylidene-1-tosyl-1,2,3,4a-tetrahydro-4H-cyclohepta[b]pyridine-4,4-dicarboxylate (**3a′**).* Yield: 0.0335 g, 71%; a colorless solid; mp: 93−98 °C; Eluent:PE/EA = 4:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 7.22 (d, J = 8.2 Hz, 2H),

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7.09 (d, J = 8.2 Hz, 2H), 6.83−6.74 (m, 2H), 6.54 (dd, J = 10.2, 5.6 Hz, 1H), 5.80 (dd, J = 8.8, 5.6 Hz, 1H), 5.64 (qd, J = 7.0, 1.4 Hz, 1H), 4.16−4.06 (m, 4H), 3.86 (t, J = 8.2 Hz, 1H), 2.77 (dd, J = 13.4, 1.4 Hz, 1H), 2.46 (dt, J = 13.4, 1.8 Hz, 1H), 2.37 (s, 3H), 1.90 (dd, J = 7.0, 1.8 Hz, 3H), 1.64 (dt, J = 7.4, 1.4 Hz, 1H), 1.21 (t, J = 7.2 Hz, 3H), 1.10 (t, J = 7.2 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 169.5, 169.0, 143.9, 134.2, 130.4, 129.2, 128.9, 128.4, 127.9, 127.0, 126.0, 118.7, 113.7, 113.0, 61.8, 61.6, 57.1, 40.3, 33.4, 21.5, 14.7, 14.1, 14.0; IR (neat): ν 2983, 2959, 2923, 2852, 1732, 1592, 1495, 1443, 1359, 1305, 1258, 1228, 1182, 1164, 1088, 1068, 1031, 921, 856, 814, 715, 667 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{25}H_{29}NO_6NaS$ 494.1608; found 494.1605.

General Procedure for the Synthesis of ***5e′***. Compound **9-(4-formylphenyl)-2-vinylcyclohepta[*b*]pyran-4,4(4a*H*)-dicarboxylate diethyl 5e** (0.042 g, 0.10 mmol) was added to an oven-dried 25 mL sealed tube equipped with a magnetic stirring bar under an argon atmosphere and added with THF (degassed) (3.0 mL). The mixture was cooled to 0 °C, and then, $LiAlH_4$ (1.0 mL, 1.0 M in THF) was slowly added. The reaction mixture was stirred at 0 °C in an ice bath for 1 h. After that, the reaction mixture was quenched with the addition of aqueous NaOH solution (1.0 M, 6.0 mL) and filtered through a pad of Celite to remove insoluble solids. The organic phases were then separated and the aqueous layer was extracted with dichloromethane (3 × 5 mL). The combined organic layers were dried over anhydrous Na_2SO_4, filtered, and evaporated. The residue was then purified by flash chromatography on silica gel (dichloromethane/methanol = 5/1) to afford the desired product **5e′** as a white solid in 72% yield (0.024 g).

*(9-(4-(Hydroxymethyl)phenyl)-2-vinyl-4,4a-dihydrocyclohepta[b]pyran-4,4-diyl)dimethanol (**5e′**).* Yield: 0.0240 g, 72%; a colorless solid; mp: 168−170 °C; Eluent: DCM/methanol = 5:1. ^1H NMR (400 MHz, Acetone-D6, TMS) δ 7.28 (d, J = 8.4 Hz, 2H), 7.22 (d, J = 8.4 Hz, 2H), 6.69−6.60 (m, 2H), 6.21 (ddt, J = 8.8, 4.0, 1.2 Hz, 1H), 6.06 (dd, J = 17.2, 10.8 Hz, 1H), 5.48 (dd, J = 8.8, 7.0 Hz, 1H), 5.20 (d, J = 1.2 Hz, 1H), 5.04 (dd, J = 17.2, 2.0 Hz, 1H), 4.83 (dd, J = 10.8, 2.0 Hz, 1H), 4.52 (d, J = 5.8 Hz, 2H), 4.07 (td, J = 5.8, 1.0 Hz, 1H), 3.90 (td, J = 5.8, 5.0 Hz, 1H), 3.82 (t, J = 2.0 Hz, 2H), 3.57 (dd, J = 10.6, 5.2 Hz, 1H), 3.37 (dd, J = 10.6, 6.2 Hz, 1H), 2.75 (s, 1H), 1.71 (dt, J = 7.0, 1.4 Hz, 1H); $^{13}C\{^1H\}$ NMR (100 MHz, Acetone-D_6, TMS) δ 148.8, 140.6, 136.5, 136.3, 130.9, 130.6, 129.9, 127.9, 126.0, 125.6, 118.2, 116.4, 113.3, 106.0, 65.6, 63.7, 62.4, 40.4, 39.4; IR (neat): ν 3345, 3022, 2922, 2870, 2851, 1660, 1597, 1461, 1407, 1375, 1336, 1282, 1234, 1217, 1036, 916, 799, 754 cm^{-1}; HRMS (ESI-FTMS) m/z: $[M + H]^+$ Calcd for $C_{21}H_{23}O_4$ 339.1591; found 339.1586.

General Procedure for the Synthesis of ***5r′***. Compound **9-chloro-2-vinylcyclohepta[*b*]pyran-4,4(4a*H*)-dicarboxylate diethyl 5r** (0.035 g, 0.10 mmol) was added to an oven-dried 25 mL sealed tube equipped with a magnetic stirring bar and added with DCM (3.0 mL). The mixture was cooled to 0 °C in an ice bath, and then, *m*-CPBA (0.021 g, 0.12 mmol) and Na_2CO_3 (0.013 g, 0.10 mmol) were added. The reaction mixture was warmed up to room temperature and stirred for 1.0 h. After that, the reaction mixture was filtered through a pad of Celite to remove insoluble solids. The solution was then purified by flash chromatography on silica gel (petroleum ether/ethyl acetate = 6/1) to afford the desired product **5r′** as a white solid in 74% yield (0.0387 g) with d.r. = 1.3:1.

*Diethyl 2-Acetyl-9-chloro-2-((3-chlorobenzoyl)oxy)-2,3-dihydrocyclohepta[b]pyran-4,4(4aH)-dicarboxylate-1 (**5r′-1**).* Yield: 0.0169 g, 32%; a colorless solid; mp: 130−134 °C; Eluent:PE/EA = 6:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 7.90 (t, J = 1.8 Hz, 1H), 7.85 (dt, J = 7.8, 1.2 Hz, 1H), 7.57 (ddd, J = 7.8, 2.2, 1.2 Hz, 1H), 7.40 (t, J = 7.8 Hz, 1H), 6.65 (d, J = 11.2 Hz, 1H), 6.58 (dd, J = 11.2, 5.2 Hz, 1H), 6.32 (ddd, J = 9.0, 5.2, 1.2 Hz, 1H), 5.11 (ddd, J = 9.0, 7.2, 0.8 Hz, 1H), 4.34−4.23 (m, 3H), 4.09−4.03 (m, 1H), 3.05 (dd, J = 15.4, 1.6 Hz, 1H), 2.78 (dt, J = 7.2, 1.6 Hz, 1H), 2.64 (d, J = 15.4 Hz, 1H), 2.41 (s, 3H), 1.24 (t, J = 7.2 Hz, 3H), 1.18 (t, J = 7.2 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 200.7, 169.4, 168.3, 163.1, 134.7, 133.9, 132.9, 130.2, 129.9, 129.9, 129.9, 128.3, 127.7, 127.0, 119.6, 110.0, 99.1, 62.5, 51.7, 36.7, 30.2, 24.6, 14.0, 13.8; IR (neat): ν 2956, 2923, 2870, 2852, 1736, 1613, 1574, 1465, 1362, 1261, 1231, 1177, 1097, 1082, 1038, 951, 746 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{25}H_{24}O_8NaCl_2$ 545.0740; found 545.0749.

*Diethyl 2-Acetyl-9-chloro-2-((3-chlorobenzoyl)oxy)-2,3-dihydrocyclohepta[b]pyran-4,4(4aH)-dicarboxylate-2 (**5r′-2**).* Yield: 0.0218 g, 42%; a colorless solid; mp: 132−136 °C; Eluent:PE/EA = 6:1. ^1H NMR (400 MHz, $CDCl_3$, TMS) δ 7.81 (t, J = 1.8 Hz, 1H), 7.75 (ddd, J = 7.8, 1.8, 1.0 Hz, 1H), 7.51 (ddd, J = 7.8, 2.2, 1.0 Hz, 1H), 7.33 (t, J = 8.0 Hz, 1H), 6.55−6.42 (m, 2H), 6.28 (dddd, J = 9.6, 4.8, 1.6, 0.8 Hz, 1H), 5.45 (dd, J = 9.0, 5.8 Hz, 1H), 4.34−4.15 (m, 4H), 3.33 (dd, J = 15.8, 2.2 Hz, 1H), 3.17 (dt, J = 5.8, 2.2 Hz, 1H), 2.87 (d, J = 15.8 Hz, 1H), 2.47 (s, 3H), 1.30−1.21 (m, 6H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, TMS) δ 202.0, 168.9, 167.7, 162.8, 136.3, 134.6, 133.8, 130.1, 130.0, 129.8, 129.7, 128.8, 128.1, 127.2, 121.2, 112.7, 99.8, 62.7, 62.6, 53.2, 38.5, 32.8, 25.1, 14.0, 13.8; IR (neat): ν 2955, 2923, 2873, 2851, 1737, 1631, 1576, 1464, 1377, 1289, 1248, 1121, 1081, 1031, 937, 859, 745 cm^{-1}; HRMS (ESI-TOF) m/z: $[M + Na]^+$ Calcd for $C_{25}H_{24}O_8NaCl_2$ 545.0740; found 545.0749.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.4c01579.

Reaction optimization, scale-up reaction, control experiments, mechanic study, asymmetric variant NMR spectra, absorption spectra, copies of ^1H NMR, $^{13}C\{^1H\}$ NMR, ^{19}F NMR for all the developed compounds, CIF files for compounds **3a**, **3ae**, **5a**, and **5s**, and computational details (PDF)

Accession Codes

CCDC 2115148, 2204243, 2255610, and 2383212 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif or by emailing data_request@ccdc.cam.ac.uk or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

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